



PANIPAT INSTITUTE OF ENGINEERING & TECHNOLOGY

(Approved by AICTE, New Delhi & Affiliated to Kurukshetra University, Kurukshetra)

AN AUTONOMOUS INSTITUTE

Panipat Institute of Engineering and Technology An Autonomous Institute Affiliated to Kurukshetra University Kurukshetra

B.Tech Computer Science Engineering Semester-V Scheme of Studies & Examination (w.e.f session 2026)

Course No.	Course Title	Contact Hrs./Sem			TW / SL	Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P				Maximum Marks	Maximum Marks	Maximum Marks	
Mandatory Credit Course											
ASH-HUM-210A	Indian knowledge System: concepts in Engineering	30	-	-	30	60	60/30=2	40	60	100	3 Hrs
Professional Core Courses											
BT-CSE-301A/BT-CYS-465A/ BT-AIML-401A/ BT-IT-301A	Theory of Computation	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-303A/ BT-AIDS-356A	Fundamentals of Artificial Intelligence	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-305A/ BT-AIDS-305A/ BT-AIML-304A/ BT IT-307A	Software Engineering	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-307A	Fundamentals of Machine Learning	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-309A/ BT-AIDS-461A/BT-IT-305A	Fundamentals of Cyber Security	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-375A	Software Engineering Lab	-	-	60	0	60	60/30=2	50	50	100	3 Hrs
BT-CSE-377A	Fundamentals Machine Learning Lab	-	-	30	0	30	30/30=1	50	50	100	3 Hrs
BT-CSE-379A	Summer Internship-II	-	-	-		-	60/30=2	50	50	100	3 Hrs
Total		255	0	90	255	600	22	390	510	900	



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HONOURS BASKET 2

BT-CSE-351A	Soft Computing	45	-	30	45	120	120/30=4	40	60	100	3 Hrs
BT-CSE-353A / BT-AIDS-355A / BT-IT-364A	Data Warehousing and Data Mining	45	-	30	45	120	120/30=4	40	60	100	3 Hrs

Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated Honours Course Basket 2.



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Course Code	Course Title	L	T	P	SL	Credits
ASH-HUM-210A	INDIAN KNOWLEDGE SYSTEM: CONCEPTS IN ENGINEERING	30	-	-	30	60/30=2
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Analyse the foundational elements of Indian Knowledge Systems and the contributions of ancient Indian universities to global knowledge development.
CO2	Understand the significance of traditional knowledge protection systems and the strategies for safeguarding indigenous intellectual property in the global economy.
CO3	Demonstrate how ancient Indian contributions to science, astronomy, and mathematics established foundational concepts that influenced global scientific development.
CO4	Analyse the technological innovations and architectural achievements of ancient Indian civilisations

Course Outcomes (CO) to Programme Outcomes (PO) mapping (Scale - 1: Low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO12	
CO1	-	-	-	-	-	3	-	2	-	-	-	3	
CO2	-	-	-	-	-	1	-	3	-	1	-	3	
CO3	1	2	-	2	-	2	-	-	-	-	-	3	
CO4	2	-	1	-	-	1	2	-	-	-	-	3	
Avg.	1.5	2	1	2	-	1.75	2	2.5	-	1	-	3	

COURSE CURRICULAM

Unit-1: Introduction to the Indian Knowledge System	Contact Hours: 6
Indian Knowledge System: The Vedas, The Vedangs, Literature (Smriti, Puran, Itihas, Upnishad), Main Schools of Philosophy, Indian Knowledge System vs Western Knowledge System; Ancient Education System: Major contributions of ancient Indian universities like Nalanda, Takshashila, Vikramshila, Odantapuri, Nagarjuna, etc., to the knowledge system; Development of Indian Education system from ancient to modern.	
Unit-2: Protection of Traditional Knowledge & Intellectual Property	Contact Hours: 6
Systems of traditional knowledge protection, The need for protecting traditional knowledge, Significance of TK Protection, the value of TK in the global economy, Role of Government to harness TK, Legal concepts for the protection of traditional knowledge, Patents and traditional knowledge, Strategies to increase protection of traditional knowledge	
Unit-3: Indian Knowledge in Science, Astronomy and Mathematics	Contact Hours: 10
Science: Concepts of Matter, Life and Universe, Gravity, Sage Agastya's Model of Battery, Velocity of Light; Vedic Cosmology, Astronomy: Bhāratīya Kāla-gaṇanā, Indian Calender system, The motion of the Sun and Moon. Mathematics: Numbers, fractions and geometry in the Vedas. Decimal nomenclature of numbers in the Vedas. Zero and Infinity. Simple constructions from Sulba-sutras, Important texts of Indian mathematics. A brief introduction to the development of algebra, trigonometry and calculus.	



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Unit-4: Indian Knowledge in Engineering, Technology, and Architecture	Contact Hours: 8
Pre-Harappan and Sindhu Valley Civilization, Laboratory and Apparatus, Juices, Dyes, Paints and Cements, Glass and Pottery, Metallurgy, Engineering Science and Technology in the Vedic Age and Post-Vedic Records, Iron Pillar of Delhi, Rakhigarhi, Mehrgarh, Sindhu Valley Civilization, Marine Technology and Bet-Dwārka	

Term Work / Self-Learning (TW/SL) Requirement

1. Assignments (Total Hours: 12 Hours)

Students must complete **four (04) assignments**, each of five (03) hours.

Suggested Assignment Areas:

- Comparative study of the Indian Knowledge System vs the Western Knowledge System.
- Contribution of ancient universities (Nalanda, Takshashila, etc.)
- Case studies on the protection of traditional knowledge (e.g., turmeric, neem patents)
- Contributions of Indian science, mathematics, and engineering

Note: Assignments will be analytical and conceptual, encouraging critical thinking rather than rote learning.

2. Activity-Based Assignments (Maximum: 10 Hours)

Students will complete up to **four (04) activity-based assignments**, each of **two and a half (02.5) hours**.

Examples: Timeline creation of Indian scientific developments, Mapping ancient texts to modern disciplines, Short analytical write-ups on philosophical schools
OR

Video-Based Learning and Report (Maximum: 10 Hours)

[Video Watching: 5 Hours + Report Writing: 5 Hours]

Students will watch curated lectures/videos on IKS topics from platforms like, SWAYAM, Mtutor, IKS India portal, and Educational YouTube channels

Submission Requirement: A detailed report including: Key learnings, Conceptual insights, Real-world relevance

3. Poster / Chart / PowerPoint Presentation (Total: 8 Hours)

Students will prepare a **poster/chart/PPT** on topics such as: Indian astronomical systems, Vedic mathematics concepts, Ancient Indian engineering achievements (Iron Pillar, Harappan drainage, etc.

Textbooks:

1. A. Jha, Traditional Knowledge System in India. New Delhi: Atlantic Publishers, 2002.
2. K. Kapoor and M. Danino, Knowledge Traditions and Practices of India. New Delhi: Central Board of Secondary Education (CBSE), 2012.
3. P. Abhang, P. Moghe, P. Holay, and S. Kulkarni, Rediscovering Indian Knowledge System. Pune: Sankalp Publication, 2022.
4. S. Raha et al., History of Science in India, vol. 1, part I, part II, vol. VIII. Kolkata: National Academy of Sciences, India and The Ramakrishna Mission Institute of Culture, 2014.
5. M. Danino, Indian Knowledge Systems: Insights for the Contemporary World. Shimla: Indian Institute of Advanced Study, 2021.
6. T. S. Bhanu Murthy, Ancient Indian Mathematics: Its Contribution to Modern Age. New



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Delhi: New Age International Publishers, 2018.

Reference Books:

1. P. Kohle et al., Eds., Pride of India: A Glimpse of India's Scientific Heritage. Delhi: Samskrita Bharati, 2006.
2. K. D. Verma, Vedic Physics. Delhi: Motilal Banarsidass Publishers, 2012.
3. S. Soni, India's Glorious Scientific Tradition. New Delhi: Ocean Books Pvt. Ltd., 2010.
4. A. L. Basham, The Wonder That Was India. London: Sidgwick & Jackson, 1954.
5. V. K. Gupta, Traditional Knowledge Digital Library: A Tool to Prevent Bio-piracy. New Delhi: Council of Scientific and Industrial Research, 2011.
6. D. F. Robinson et al., Protecting Traditional Knowledge: The WIPO Intergovernmental Committee on Intellectual Property and Genetic Resources, Traditional Knowledge and Folklore. Geneva: World Intellectual Property Organization (WIPO), 2017.
7. S. Mohan, Indian Science Through the Ages. New Delhi: Publications Division, Ministry of Information and Broadcasting, Government of India, 2015.
8. D. Chattopadhyaya, Science and Civilization in Ancient India. Calcutta: Firma KLM, 1977.

E-Resources:

1. https://onlinecourses.swayam2.ac.in/imb23_mg53/preview
2. https://onlinecourses.swayam2.ac.in/ntr25_ed18/preview



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Course Code					Course Title					L	T	P	SL	Credits
BT-CSE-301A / BT-CYS-465A/ BT-AIML-401A / BT-IT-301A					Theory of Computation					45	0	0	45	90/30 = 3
										CIE		SEE		Total
										40		60		100
Course Outcomes: After Completion of Course, Students will be able to														
CO1	Understand the fundamental concepts of automata theory and formal languages.													
CO2	Design regular and context-free grammar for formal languages and automata with rigorously formal mathematical method.													
CO3	Develop and analyze context-free grammar, normal forms, and pushdown automata, and evaluate their applications and limitations in language processing.													
CO4	Design Turing Machines, analyze decidability and computability, and apply concepts such as PCP, Rice’s theorem, and NP-completeness to classify computational problems.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	
CO1	3	3	2	1	2	–	–	1	1	–	2	2	2	
CO2	3	3	2	2	2	–	–	1	1	–	2	2	2	
CO3	3	3	3	2	3	–	–	1	1	–	2	2	3	
CO4	3	3	3	3	3	–	–	1	2	1	3	3	3	
Avg.	3	3	2.5	2	2.5	0	0	1	1.25	0.25	2.25	2.25	2.5	

Course Outline

Unit-1: Introduction	Contact Hours: 12
Basic Concepts and Automata Theory: Introduction to Theory of Computation- Automata, Computability and Complexity, Applications of Finite Automata, An Introduction of Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NFA), Finite Automata with Epsilon (ε) Transitions. Equivalence and Minimization of NFA and DFA automata. Finite Automata with output: Moore Machine, Mealy Machine, Equivalence of Moore and Mealey Machines and its Designing.	
Unit-2: Regular Expression	Contact Hours: 12
Regular Expression and Languages: Regular Expressions (RE), Finite Automata and Regular Expressions, Applications of Regular Expressions, Algebraic Laws of Regular Expressions, Closure Properties of Regular Languages, RE to NFA, DFA Conversion and DFA to RE, Finite Automata and Regular Expression- Arden's theorem, Algebraic Method Using Arden's Theorem Pumping Lemma: Introduction to Pumping Lemma Application of Pumping Lemma, Decidability- Decision properties, Finite Automata and Regular Languages	



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Unit-3: Context free Grammars and Languages	Contact Hours: 11		
Context free Grammars and Languages: Parse Trees, Context Sensitive Grammar, Context Free Grammar, Regular Grammar, Applications of Context Free Grammars, Ambiguity in Grammars and Languages. Closure Properties of CFL, Chomsky Theorem, Chomsky Hierarchy, Normal forms of context free grammars: Chomsky Normal Form, Greibach Normal Form. Push Down Automata: Introduction of Push Down Automata (PDA), Language of PDA, Equivalence of PDA's and CFG's, Deterministic Push Down Automata, Designing of PDA, Applications			
Unit-4: Turing Machines and Decidability	Contact Hours: 10		
Introduction to Turing Machine: The Turing Machine, Programming Techniques for Turing Machine, Extensions of Turing Machine, Restricted Turing Machines, Universal Turing Machines and Designing of Turing Machines, Time and Tape Complexity Measures of Turing machines. Decidability: Post's Correspondence Problem (PCP), Rice's Theorem, Decidability and Undecidability properties-NP class and completeness.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	5 assignments	20	Based on the assignments submitted.
Complex Problem Solving	2 Complex Problem (Design & prove a computational model)	10	Based on the depth and quality of the solution submitted.
Technical Video Based Learning	Related to Theory of Computation / Automata	10	Based on Summary Report
Online Technical Quizzes	5 Online Quizzes	5	Based on quiz scores and reflection reports.
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam/Oracle Academy (As Per Available Hours/Credit's). - Certificate & Report Mandatory Before End of Semester. If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. J.E.Hopcroft, R.Motwani and J.D.Ullman , "Introduction to Automata Theory Languages and computation", Pearson Education Asia, 2001.
2. K.Krithivasan and R.Rama; Introduction to Formal Languages, Automata Theory and Computation; Pearson Education, 2009.
3. Peter Linz, "An Introduction to Formal Language and Automata", 4th Edition, Narosa Publishing house, 2006.
4. M.Sipser; Introduction to the Theory of Computation; Singapore: Brooks/Cole, Thomson Learning,



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1997.

5. John.C.martin, "Introduction to the Languages and the Theory of Computation", Third edition, Tata McGrawHill, 2003

Reference Books:

1. Harry R. Lewis and Christos H. Papadimitriou, Elements of the Theory of Computation, Prentice Hall, 2nd Edition, 1998.
2. Shyamalendu Kandar, Introduction to Automata Theory, Formal Languages and Computation, Pearson Education, 2012.
3. V. R. Narasimhan, Theory of Automata, Formal Languages and Computation, New Age International Publishers, 2009.
4. Sanjiv Kapoor and Sanjay Kumar, Introduction to Automata Theory and Formal Languages Cambridge University Press India, 2019.

E-Resource:

1. <https://archive.nptel.ac.in/courses/106/106/106106049/>

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**
2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



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Course Code		Course Title								L	T	P	SL	Credits	
BT-CSE 303 A BT-AIDS-356A		Fundamentals of Artificial Intelligence								45	0	0	45	90/30=3	
										CIE		SEE		Total	
										40		60		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Understand the fundamental concepts of AI, intelligent agents, and problem formulation and Industry 4.0 applications.														
CO2	Apply search techniques and optimization methods to solve AI problems. .														
CO3	Analyze knowledge representation methods and reasoning under uncertainty. .														
CO4	Explore AI planning, AI tools, ethical AI concepts, and real-world applications of Artificial Intelligence.														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	3	2	-	-	2	-	-	-	-	-	-	2	1		
CO2	3	3	2	2	3	-	-	-	1	-	-	3	2		
CO3	3	3	2	3	2	-	-	-	1	-	-	3	2		
CO4	2	3	3	2	3	1	1	1	2	1	-	2	3		
Avg.	2.75	2.75	2.33	2.33	2.50	1.00	1.00	1.00	1.33	1.00	-	2.50	2.00		

Course Outline

Unit-1: Foundations of AI, Intelligent Agents & Problem Formulation	Contact Hours: 11
Introduction to Artificial Intelligence, History, Evolution & Foundations of AI, Types of AI, Applications of AI, AI in Automation & Industry 4.0, Intelligent Agents and Agent Architecture, Types of Agents: Simple Reflex, Model-based, Goal-based, Utility-based, PEAS Frameworks: Performance, Environment, Actuators, Sensors, Problem Formulation in AI, State-Space Representation, Production Systems, Problem Formulation in Data-Driven.	
Unit-2: Problem Solving, Search Strategies & Optimization	Contact Hours: 11
Production systems, Problem space and state space search, Graph search strategies: Uninformed search (Depth-first search and Breadth-first search, Uninformed cost search, Depth Limited Search, Iterative Deepening Search), Heuristic search (Hill climbing, Best-first search, A* algorithm and AO* algorithm), Adversarial search, Min-Max algorithm, Alpha-beta pruning, General search and optimization search, branch and bound.	
Unit-3: Knowledge Representation & Reasoning	Contact Hours: 11
Knowledge Representation Models, Knowledge-Based Systems, Propositional Logic, Predicate / First Order Logic, WFF, Quantifiers, Syntax & Semantics, Inference Mechanisms: Forward Chaining, Backward Chaining, Resolution, Unification, Structured Representations: Semantic Networks, Frames, Scripts, Ontologies, Description Logic, Fuzzy Logic, Fuzzy Inference	



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Unit-4: AI Planning, Tools and Applications			Contact Hours: 12
AI Planning Fundamentals, Problem Formulation for Planning, State-Space Planning, Goal Stack Planning, Partial Order Planning, Planning vs Acting, Multi-Agent Planning, AI tools and platforms: OpenAI ChatGPT, Google Gemini, Microsoft Copilot, TensorFlow, PyTorch, OpenCV, Scikit-learn and Keras. Applications of AI in healthcare, finance, robotics, cybersecurity, smart systems, autonomous systems and recommendation systems, and ethical and responsible AI systems: fairness, bias and explainable AI.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	5 assignments Unit Wise	20	Based on the assignments submitted.
Problem Solving / Coding	5 Python problems (Related to Subject)	05	Based on the coding solutions submitted.
Real-World Case Study-Based Learning	Case study on Tools and Applications	10	Based on technical depth, data collection, analysis and findings.
Micro Project	Related To AI	10	Based on Project Report Submitted.
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's). - Certificate & Report Mandatory Before End of Semester. If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. E. Rich and K. Knight, "Artificial Intelligence", McGraw Hill Education; 3rd edition, 2017.
2. Luger F. G., Artificial Intelligence: Structures and Strategies for Complex Problem Solving, Pearson Education Asia (2009) 6th ed.

Reference Books:

1. Patterson W. D., Introduction to Artificial Intelligence and Expert Systems, Pearson (2015) 1st ed.
2. Russel S., Norvig P., Artificial Intelligence: A Modern Approach, Prentice Hall (2014) 3rd ed.
3. N. J. Nilsson, "Principles of AI", Narosa Publ. House, 1990.
4. Artificial intelligence, Patrick Henry Winston: 1992, Addition Wesley 3Ed.
5. P. H. Winston, "Artificial Intelligence", Pearson Education, 3rd Edition, 2002.
6. R. J. Schalkoff, "Artificial Intelligence – An Engineering Approach", McGraw Hill Int. Ed. Singapore, 1992.

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall



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be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. All questions shall carry equal marks.

2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



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Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-305A / BT-AIDS-305A / BT-AIML-304A / BT-IT-307A	Software Engineering	45	0	0	45	90/30=3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Apply the concepts to choose an appropriate software process model according to given user requirements.
CO2	Analyze requirement techniques like Data flow diagram, Entity relationship diagram, Object diagram.
CO3	Learn software design concepts, explore various design strategies, and understand metrics to assess and enhance software quality.
CO4	Apply testing techniques and reliability measures to assess software quality and performance under faults and failures.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	1	-	-	-	1	-	1	2	2
CO2	2	2	2	2	1	-	-	-	1	-	1	1	2
CO3	2	2	3	2	2	-	-	-	1	-	1	2	3
CO4	2	2	3	2	2	-	-	-	1	-	2	2	2
Avg	2.25	2	2.5	1.75	1.5	-	-	-	1	-	1.25	1.75	2.25

Course Outline

Unit-1: Software Processes and Metrics	Contact Hours: 11
Introduction: Software Crisis, software processes and its models (waterfall, incremental development, spiral model, re-use-oriented model, prototype), Process activities, Process improvement (CMM Levels). Overview of Quality Standards like ISO 9001, SEI-CMM. Software Metrics: Size Metrics like LOC, Token Count, Function Count, Design Metrics, Data Structure Metrics, Information Flow Metrics.	
Unit-2: Requirement Engineering	Contact Hours: 12
Requirement Engineering: Functional and non-functional requirements, requirement elicitation, use case development, requirement analysis and validation, requirement review or requirement change, SRS document. Size Estimation: Software Size, LOC and function point, cost and effort estimation, COCOMO, ISO 9001:2015 Certification. System modelling: Interaction models: Use case diagram, Structural models: class diagrams, generalization, aggregation, Behavioural models: ER diagrams, Data flow diagrams, data dictionaries.	
Unit-3: Software Design	Contact Hours: 12
Software Design: Architectural views and patterns, Modularity (cohesion and coupling), information hiding, functional independence, function-oriented design, object-oriented design, SOA, SAAS. Software Quality: McCall's Quality Factors, ISO 9126 Quality Factors, Quality Control, Quality Assurance, Software Reliability. Software Evolution: Evolution process, legacy system, software maintenance: Maintenance prediction, Re Engineering, Refactoring.	



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Unit-4: Software Testing			Contact Hours: 10
Software Testing: verification, validation, Development testing (unit testing, component testing, system testing, Test Driven Development (TDD), Release Testing (Requirement based testing, scenario testing, performance testing), User testing (alpha, beta and acceptance testing), Regression Testing, Stress Testing. System Security: Reliability engineering, reliability requirements (functional and non-functional) and its measurement, safety engineering: safety critical systems, its requirements, security engineering and its requirements, security guidelines.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	5 Assignments	20	Based on the assignments submitted.
Real-World Case Study- Based Learning	Case study on a software project / failure analysis	10	Based on technical depth, data collection, analysis and findings.
Online Technical Quizzes	5 quizzes on Unit Wise/Complete Syllabus	5	Based on quiz scores and reflection reports.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	10	Based on poster / chart quality and presentation skills With Summary Report
OR			
• Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's). • Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Ian Sommerville, "Software Engineering", 10th edition, Pearson, 2018.
2. K.K. Aggarwal & Yogesh Singh, "Software Engineering", New Age International Publishers, New Delhi, Third Edition, 2008.

Reference Books:

1. Pankaj Jalote, "A Concise Introduction to Software Engineering", Springer, 2008.
2. Roger S. Pressman, "Software Engineering- A Practitioner's Approach", Eighth Edition, McGraw- Hill International Edition, 2010.
3. Cem Kaner, Jack Falk, Nguyen Quoc, "Testing Computer Software", Second Edition, Van Nostrand Reinhold, New York, 1993.

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions**



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shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**

2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



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Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-307A	Fundamentals of Machine Learning	45	0	0	45	3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Understand ML fundamentals, mathematics, and data preprocessing.
CO2	Apply supervised, unsupervised, and basic deep learning algorithms.
CO3	Evaluate ML models using metrics, validation, and ethical principles.
CO4	Create basic ML pipelines incorporating deployment, MLOps, and modern AI techniques.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	1	2	-	-	-	1	-	2	3	2
CO2	3	3	2	2	3	-	-	1	1	-	2	3	3
CO3	2	3	1	2	2	2	3	-	1	-	2	3	2
CO4	3	3	3	2	3	-	-	2	2	2	3	3	2
Avg	2	2	2.25	2.25	2	2	3	1.5	1.5	1.5	1	3	3

Course Outline

Unit-1: Machine Learning Foundations & Data Preprocessing	Contact Hours: 12
Introduction : Evolution of ML, Types of ML: Supervised, Unsupervised, Reinforcement Learning, ML workflow and industry use cases (e-commerce, healthcare, fintech, autonomous systems), Mathematical Foundations: Linear Algebra basics (Vectors, Matrices, Eigenvalues, Eigenvectors), Probability & Statistics (Mean, Variance, PDF, CDF, Bayes Theorem), Calculus essentials (Derivatives, Gradients, Partial Derivatives), Data Preprocessing: Handling missing data, outliers, and duplicates, Feature scaling: Normalization vs Standardization, Encoding categorical data, Data transformation pipelines in Python (scikit-learn, pandas) Industry Tools & Practices: Data collection techniques, Data versioning with DVC.	
Unit-2: Core ML Algorithms & Evaluation	Contact Hours: 12
Supervised Learning: Linear Regression & Ridge/Lasso Regularization, Logistic Regression, Decision Trees and Random Forests, Support Vector Machines (SVM), Unsupervised Learning: Clustering: K-Means, Hierarchical Clustering, Dimensionality Reduction: PCA, t-SNE, Model Evaluation: Train-Test Split & Cross-validation, Metrics: Accuracy, Precision, Recall, F1-score, AUC-ROC, Bias-Variance Trade-off, Deployment Concepts: Saving and loading models (Pickle, Joblib), Introduction to ML pipelines.	
Unit-3: Advanced ML, Deep Learning & Industry Frameworks	Contact Hours: 11
Introduction to Neural Networks: Perceptrons, activation functions, Gradient Descent and Backpropagation, Deep Learning Architectures: Convolutional Neural Networks (CNNs) for image processing, Recurrent Neural Networks (RNNs) and LSTMs for sequence data, Transformers overview (real-world applications like ChatGPT, BERT), Industry-Standard Frameworks: TensorFlow basics, PyTorch basics, Introduction to cloud-based ML (Google Colab, AWS SageMaker, Azure ML).	



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Unit-4: Industry Applications, Ethics & MLOps			Contact Hours: 10
Real-World Applications: Healthcare (disease prediction, medical imaging) , Finance (fraud detection, credit scoring), Retail & E-commerce (recommendation engines, demand forecasting) , Autonomous systems (self-driving cars, robotics), MLOps & Production Readiness: Continuous Integration/Continuous Deployment (CI/CD) for ML, Model monitoring and retraining, Tools: MLflow, Docker, Kubernetes basics for ML Ethics and Responsible AI: Fairness, accountability, transparency, Bias in ML models, GDPR and data privacy laws, Industry Trends: Generative AI and LLMs, Quantum Machine Learning, AutoML and zero-code ML platforms.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	5 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 Python (scikit-learn) model-building tasks	10	Based on the coding solutions submitted.
Online Technical Quizzes	5 quizzes on Unit Wise/Complete Syllabus	5	Based on quiz scores and reflection reports.
Micro Project	Related To ML	10	Based on Project Report Submitted.
OR			
• Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's). • Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Publisher: O'Reilly Media, Year: 2019 (2nd Edition)
2. François Chollet, Deep Learning with Python, Publisher: Manning Publications, Year: 2017 (1st Edition) / 2021 (2nd Edition)
3. Christopher M. Bishop, Pattern Recognition and Machine Learning, Publisher: Springer, Year: 2006
4. Sebastian Raschka & Vahid Mirjalili, Python Machine Learning, Publisher: Packt Publishing, Year: 2019 (3rd Edition)
5. Tom M. Mitchell, Machine Learning, Publisher: McGraw-Hill, Year: 1997

Reference Books:

1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, Publisher: MIT Press, Year: 2016
2. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Publisher: Springer, Year: 2009 (2nd Edition)
3. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, Publisher: MIT Press, Year: 2012
4. Ethem Alpaydin, Introduction to Machine Learning, Publisher: MIT Press, Year: 2020 (4th Edition)
5. Andrew Ng, Machine Learning Yearning, Publisher: deeplearning.ai, Year: 2018



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Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**
2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



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Course Code				Course Title						L	T	P	SL	Credits	
BT-CSE-309A / BT- AIDS-461A / BT-IT-305A				Fundamentals of Cyber Security						45	0	0	45	90/30=3	
										CIE		SEE		Total	
										40		60		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Understand cybercrime and its laws.														
CO2	Analysis tools and methods used in cybercrime.														
CO3	Apply computer forensics methods to analyze digital evidence.														
CO4	Understand web threats, challenges and protection policies.														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	1	1	–	–	–	3	–	3	–	2	1	–	1		
CO2	2	2	1	2	3	1	–	1	–	2	2	1	2		
CO3	2	3	–	3	3	1	–	2	–	2	2	1	2		
CO4	2	2	2	2	3	3	–	3	2	3	2	1	2		
Avg	1.75	2	1	1.75	3	2	–	2.25	0.5	2.25	1.75	0.75	1.75		

Course Outline

Unit-1: Introduction To Cyber Crime	Contact Hours: 12
Cybercrime and Information Security, Classifications of Cybercrimes, The need for Cyber laws, The Indian IT Act 2000 Challenges to Indian Law and Cybercrime Scenario in India, Weakness of IT Attacks and its consequences, Digital Signatures, Cybercrime and Punishment, Survival tactics for the Citizens, Cyber-offenses: Cybers Stalking, Types of Cybercrimes, Attack Vector.	
Unit-2: Tools And Methods Used in Cybercrime	Contact Hours: 11
Proxy Servers and Anonymizers, Phishing and identity theft, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer Overflow, Cybercrime Mobile and Wireless Devices Trends in Mobility, Attacks on Wireless Networks, Credit Card Frauds in Mobile and Wireless Computing Era, Security Challenges, Registry Settings for Mobile Devices, Authentication Service Security Attacks on Mobile/Cell Phones.	
Unit-3: Understanding Computer Forensics	Contact Hours: 10
The Need for Computer Forensics, Cyber forensics and Digital Evidence, Forensics Analysis of E-Mail, Digital Forensics Life Cycle, Chain of Custody Concept, Network Forensics, Computer Forensics and Stenography, Relevance of the OSI Layer Model to Computer Forensics, Forensics and Social Networking Sites The Security/Privacy Threats, Challenges in Computer Forensics, Forensics Auditing, Anti forensics.	



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Unit-4: Organizational Implications			Contact Hours: 12
Cost of Cybercrimes and IPR Issues, Web Threats for Organizations, Security and Privacy Implications from Cloud Computing, Social Media Marketing, Social Computing and the Associated Challenges for Organizations, Protecting People's Privacy in the Organization, Organizational Guidelines for Internet Usage, Safe Computing Guidelines and Computer Usage Policy, Incident Handling, Forensics Best Practices, Media and Asset Protection, Importance of Endpoint Security in Organizations.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	5 Assignments	20	Based on the assignments submitted.
Real-World Case Study-Based Learning	Case study on a real cyber-attack / breach	10	Based on technical depth, data collection, analysis and findings.
Video Based Learning	Video Based on Cyber Security fundamentals/Attacks	10	Based On Summary Report
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's). - Certificate & Report Mandatory Before End of Semester. If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks

1. Nina Godbole and Sunit Belapure, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", 2nd Edition, Wiley India, 2011.
2. Nelson, Phillips and Steuart, "Guide to Computer Forensics and Investigations", 6th Edition, Cengage Learning, 2019.
3. William Stallings, "Computer Security: Principles and Practice", 4th Edition, Pearson Education, 2018.

Reference Books

1. Douglas J. Landoll, "The Security Risk Assessment Handbook", 3rd Edition, CRC Press, 2016.
2. Bill Nelson, Christopher Steuart, "Computer Forensics and Cyber Crime: An Introduction", 3rd Edition, Cengage Learning, 2014.
3. Pavan Duggal, "Cyber Security Law", 1st Edition, LexisNexis, 2019.
4. Chuck Easttom, "Computer Security Fundamentals", 4th Edition, Pearson Education, 2019.
5. Kevin Mandia, Chris Prosise, "Incident Response and Computer Forensics", 2nd Edition, McGraw Hill, 2003.
6. James R. Aquilina, Eoghan Casey, "Handbook of Digital Forensics and Investigation", Academic Press, 1st Edition, 2010.



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Instructions for Semester-End Examination

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Course Code				Course Title				L	T	P	SL	Credits	
BT-CSE-375A				Software Engineering Lab				-	-	60	0	60/30=2	
								CIE		SEE		Total	
								50		50		100	
Course Outcomes: After Completion of Course, Students will be able to													
CO1		Translate end-user requirements into system and software requirements											
CO2		Generate a high-level design of the system from the software requirements											
CO3		Analysis the testing problems and will be able to develop a simple testing report											
CO4		Apply project management principles to plan, schedule, and manage software development activities using Gantt/PERT charts.											
Course Outcomes (CO) to Program Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)													
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	–	–	–	–	–	–	2	–	1	1
CO2	2	3	3	–	3	–	–	–	–	2	–	1	1
CO3	3	2	3	2	2	–	–	–	–	–	–	2	2
CO4	–	–	2	–	2	–	–	–	3	3	3	2	2
AVG	2	2	2.5	0.5	1.75	–	–	–	3	2.33	3	1.5	1.5

List of Experiment

1. Perform Problem Identification and Feasibility Analysis for a Software System.
2. Develop a Software Requirement Specification (SRS) Document for the Selected System.
3. Design Data Flow Diagrams (DFD) for the Selected Software System.
4. Design a Structure Chart for the Selected Software System.
5. Develop a Use Case Diagram for the Selected Software System.
6. Develop Class Diagram and Object Diagram for the Selected Software System.
7. Develop Activity Diagram and State Chart Diagram for the Selected Software System.
8. Develop Sequence Diagram and Communication Diagram for the Selected Software System.
9. Design a Component Diagram for the Selected Software System.
10. Design a Deployment Diagram for the Selected Software System.
11. Prepare Work Breakdown Structure (WBS), Gantt Chart, and PERT Chart for the Selected Software Project.
12. Perform Software Cost Estimation and Risk Analysis for the Selected Software Project.
13. Design and Execute Unit Test Cases for the Selected Software System.
14. Perform Integration Testing and System Testing for the Selected Software System.
15. Implement Version Control and Software Configuration Management using Git/GitHub.

Textbooks:



1. Terry Quatrani, Visual Modeling with Rational Rose and UML, Addison-Wesley, 3rd Edition, 2000
2. Richard C. Lee and William M. Tepfenhart, Mastering UML with Rational Rose 2002, Pearson / Prentice Hall, 2002

Reference Books:

1. Pascal Roques, Mastering UML with Rational Rose 2002, Sybex / Wiley, 2002.
2. Jim Arlow & Ila Neustadt, UML and the Unified Process: Practical Object-Oriented Analysis and Design, Addison-Wesley, 2002.
3. Srinivasan Desikan, Gopalswamy Ramesh, Software Testing: Principles and Practices, Pearson Education, 2006.
4. Roger S. Pressman, Software Engineering: A Practitioner's Approach, McGraw-Hill, 2010.

NOTE: A student must perform at least Fifteen experiments. Twelve experiments should be performed from the above list. The remaining Three experiments may either be performed from the above list or as per the scope of the syllabus.



Course Code	Course Title	L	T	P	SL	Credits							
BT-CSE-377A	Fundamental of Machine Learning Lab	-	-	30	0	30/30=1							
		CIE	SEE	Total									
		50	50	100									
Course Outcomes: After Completion of Course, Students will be able to													
CO1	Implement supervised and unsupervised machine learning algorithms using Python and related libraries.												
CO2	Perform data preprocessing, visualization, and feature selection for various datasets.												
CO3	Analyze model performance using appropriate evaluation metrics and tuning techniques.												
CO4	Apply deep learning frameworks for solving NLP and computer vision related tasks.												
Course Outcomes (CO) to Program Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)													
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	1	2	3	–	–	–	1	1	–	2	3
CO2	2	2	1	2	3	–	–	–	1	1	–	2	3
CO3	2	3	1	3	3	–	–	–	1	2	1	2	3
CO4	2	2	2	2	3	–	–	–	1	2	1	2	3
Avg.	2.00	2.25	1.25	2.25	3.00	–	–	–	1.00	1.50	0.50	2.00	3.00

List of Experiments:

1. Perform Installation and Configuration of Python, NumPy, Pandas, Matplotlib, and Scikit-learn for Machine Learning Applications.
2. Perform Data Preprocessing and Visualization using Scikit-learn and Seaborn.
3. Implement Linear Regression and Polynomial Regression Models for Predictive Analysis.
4. Implement Decision Tree Classification and Visualize the Decision Tree Structure.
5. Implement K-Nearest Neighbors (KNN) Algorithm and PCA-Based Dimensionality Reduction.
6. Implement Naïve Bayes Classifier and Logistic Regression on Benchmark Datasets.
7. Implement Support Vector Machine (SVM) using Linear and Kernel-Based Approaches.
8. Implement Ensemble Learning Techniques using Bagging, Boosting, and Random Forest Algorithms.
9. Implement K-Means Clustering Algorithm for Unsupervised Learning Applications.
10. Implement Hierarchical Clustering and Analyze Cluster Formation.
11. Implement Artificial Neural Networks (ANN) using TensorFlow/Keras Framework.
12. Implement Deep Learning Models for Image Classification using MNIST/CIFAR-10 Datasets.
13. Implement Natural Language Processing Applications using RNN/LSTM for Text Classification.
14. Perform Comparative Analysis of Machine Learning Models using Appropriate Performance Evaluation Metrics.



15. Develop a Mini Project by Applying Machine Learning Techniques on a Real-World Dataset.

Text Books

1. Tom M. Mitchell, Machine Learning, McGraw-Hill Education, 1st Edition, 1997. ISBN: 978-0070428072.
2. Ethem Alpaydin, Introduction to Machine Learning, MIT Press, 4th Edition, 2020. ISBN: 978-0262043793.
3. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012. ISBN: 978-0262018029.
4. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006. ISBN: 978-0387310732.

Reference Books

1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016. ISBN: 978-0262035613.
2. Charu C. Aggarwal, Neural Networks and Deep Learning, Springer, 2nd Edition, 2023. ISBN: 978-3031467219.
3. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer, 2nd Edition, 2009. ISBN: 978-0387848570.

NOTE: A student must perform at least Fifteen experiments. Twelve experiments should be performed from the above list. The remaining Three experiments may either be performed from the above list or as per the scope of the syllabus.



Course Code				Course Title					L	T	P	Credits		
BT-CSE-379A				Summer Internship -II					-	-	-	2		
									CIE		SEE		Total	
									50		50		100	
Course Outcomes: After Completion of Course, Students will be able to														
CO1	Apply foundational technical knowledge and programming skills to practical tasks and simple real-world projects.													
CO2	Understand and use laboratory tools, departmental resources, or institutional facilities to gain hands-on exposure.													
CO3	Develop documentation, reporting, and basic communication skills through daily logs and internship reports													
CO4	Demonstrate professional ethics, teamwork, time management, and problem- solving abilities in real-world or institutional activities.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	
CO1	3	3	2	2	2	2	–	–	1	–	2	3	2	
CO2	2	3	2	2	2	2	–	–	2	–	2	2	3	
CO3	2	2	2	2	2	–	–	3	3	2	2	1	2	
CO4	2	2	1	2	2	–	–	3	3	2	3	2	2	
AVG	2.5	2.75	2	2.33	2.67	1	–	–	1.5	1	2	2.75	3	

B.Tech students are required to attend Inter/Intra Institute Summer Internship-II of 4–6 weeks duration at the end of the 4th semester (before the commencement of 5th semester) with the following objectives:

- Provide opportunities to learn, understand, and enhance foundational technical and managerial skills relevant to the student's discipline.
- Expose students to current technological developments and practical applications related to their field of study.

Students are expected to be involved in one or more of the following activities, as recommended by the Departmental Internship Coordinator and approved by the Head of Department:

- Training with higher institutions or industry.
- Soft skill development programs organized by the Training and Placement Cell.
- Contribution to Incubation/Innovation/Entrepreneurship Cell.
- Participation in conferences, workshops, competitions, or technical expos.
- Learning at departmental labs, tinkering labs, or institutional workshops.
- Working on departmental or institutional research/consultancy projects.
- Participation in activities organized by the Institute's Innovation Council, e.g., IPR workshops, leadership talks, idea/design/innovation/business competitions, and technical expos.

Documentation Requirements:

- Students are required to maintain a daily diary to cultivate the habit of documenting observations and learning experiences.



- At the end of the internship, students shall prepare a comprehensive report highlighting observations, tasks performed, and knowledge gained.
- Students may consult the Internship Coordinator or Faculty Mentor for special topics or problems and should include findings in the final report.
- The daily diary will serve as a key resource for writing the final internship report.

Evaluation of Internship Report and Presentation:

The internship report will be evaluated based on:

1. Originality (a signed certificate of originality must be attached).
2. Adequacy and purposefulness of the write-up.
3. Organization, format, diagrams/sketches, style, language, and variety of learning experiences.
4. Practical application and linkage to theoretical concepts taught in the curriculum.

Presentation Evaluation Committee:

- Head of Department
- Internship Coordinator
- Class In-Charge



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(5th Sem) Honours Course (Basket-2)



Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-351A	Soft Computing	45		30	45	120/30=4
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Explain and analyze soft computing techniques and their applications in building intelligent systems.
CO2	Analyse fuzzy sets, membership functions, and fuzzy inference systems for decision-making.
CO3	Analyse neural networks and optimization techniques for machine learning.
CO4	Analyse genetic algorithms for optimization and machine learning.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	2	2	–	–	–	1	1	–	2	1
CO2	3	3	2	2	3	–	–	–	1	1	–	2	2
CO3	3	3	3	2	3	–	–	–	2	1	1	3	2
CO4	3	3	3	2	3	–	–	–	2	1	1	2	3
Avg.	3.00	2.75	2.50	2.00	2.75	–	–	–	1.50	1.00	0.50	2.25	2.00

Course Outline

Unit-1: Introduction	Contact Hours: 10
Soft Computing and Neural Networks: Evolution of Computing & need for soft computing, Soft vs Hard Computing; From Conventional AI to Computational Intelligence, Soft Computing Constituents (FL, ANN, GA, PR and their hybrids), Machine Learning Basics : types of learning, generalization, bias-variance trade-off, Foundations of Neural Networks : biological neuron, McCulloch-Pitts model, activation functions.	
Unit-2: Fuzzy Logic	Contact Hours: 12
Overview of Classical (Crisp) Sets and Fuzzy Sets, Membership Functions : types and features, Operations on Fuzzy Sets :Complement, Intersection, Union, Combinations of Operations, Aggregation Operations; Fuzzy Relations and their Composition; Fuzzy Rules, Fuzzy Rule Generation and Fuzzy Reasoning; Fuzzy Inference Systems : Mamdani v/s Sugeno models, Fuzzy Expert Systems, Fuzzy Decision Making.	
Unit-3: Neural Networks	Contact Hours: 11
Machine Learning using Neural Networks, Feed-forward Networks, Supervised Learning Neural Networks (perceptron, back-propagation), Radial Basis Function (RBF) Networks, Unsupervised Learning Neural Networks, Reinforcement Learning, Adaptive Networks, Adaptive Resonance Theory (ART) architectures, Advances in Neural Networks, Introduction to swarm-based learning : Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO).	



Unit-4: Nature Inspired Algorithm			Contact Hours: 12
Optimization problems and traditional optimization methods, Limitations of traditional methods and differences between Genetic Algorithms (GA) and traditional methods, Introduction to Genetic Algorithms and their Terminology; Simple Genetic Algorithm (SGA), Different types of GA operators, Selection, Crossover, Mutation, and Replacement, Applications of GA in Machine Learning, Machine Learning Approach to Knowledge Acquisition			
List of Experiment			
Unit-1: Introduction to Soft Computing & Neural Networks			Contact Hours: 6
1. Study and comparison of Soft Computing vs Hard Computing approaches 2. Implementation of basic Machine Learning workflow 3. Study of Conventional AI vs Computational Intelligence			
Unit-2: Fuzzy Logic			Contact Hours: 8
4. Implementation of Fuzzy Sets and Membership Functions 5. Operations on Fuzzy Sets (Union, Intersection, Complement, Aggregation) 6. Design of a Fuzzy Inference System using Mamdani or Sugeno method			
Unit-3: Neural Networks			Contact Hours: 8
7. Implementation of Feed Forward Neural Network using supervised learning 8. Implementation of Unsupervised Learning Neural Network (SOM / ART) 9. Implementation of Radial Basis Function Neural Network			
Unit-4: Nature Inspired Algorithm			Contact Hours: 8
10. Implementation of a Simple Genetic Algorithm (SGA) 11. Study of Genetic Algorithm Operators (Selection, Crossover, Mutation). 12. Application of Genetic Algorithm in Machine Learning Optimization			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	5 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 Python soft-computing problems	10	Based on the coding solutions submitted.
Video Based Learning	Video Based on Soft Computing	10	Based On Summary Report
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's). - Certificate & Report Mandatory Before End of Semester. If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			



Textbooks:

1. S. N. Sivanandam & S. N. Deepa, Principles of Soft Computing, Wiley India Pvt. Ltd.
2. Goldberg D. E., Genetic Algorithms in Search, Optimization, and Machine Learning, Pearson Education.
3. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron
4. Deep Learning with Python – François Chollet

Reference Books:

1. Jang, Sun, Mizutani, Neuro-Fuzzy and Soft computing, Pearson Education.
2. Haykin, Neural networks: a comprehensive foundation, Pearson Education.
3. George J. Klir and Bo Yuan, “Fuzzy Sets and Fuzzy Logic: Theory and Applications”, Prentice Hall, 1995.
4. Make Your Own Neural Network – Tariq Rashid

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**
2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



Course Code				Course Title						L	T	P	SL	Credits	
BT-CSE-353A / BT-AIDS-355A / BT-IT-364A				Data Warehousing and Data Mining						45		30	45	120/30=4	
										CIE		SEE		Total	
										40		60		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Understand and explain the principles and concepts of Data Warehousing and Data Mining.														
CO2	Analyze the architecture and components of a Data Mining system.														
CO3	Apply data mining techniques such as classification, association, and prediction to real datasets.														
CO4	Explore and evaluate emerging trends in data mining, web mining and spatial-temporal mining.														
Course Outcomes (CO) to Programme Outcomes (PO) and PSOs mapping (scale 1: low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	3	2	2	3	2	–	2	–	–	–	–	2	2		
CO2	2	2	3	2	2	–	2	–	–	–	–	1	2		
CO3	3	3	2	2	3	–	–	–	–	–	–	2	3		
CO4	2	2	3	2	3	–	–	2	–	–	1	2	2		
Avg.	2.50	2.25	2.50	2.25	2.50	–	1.00	0.50	–	–	0.25	1.75	2.25		

Course Outline

Unit: I	Data Warehousing – Introduction and Design:	Contact Hours:12
Overview and Concepts: Data Warehousing Components, Building a Data Warehouse, Data Warehouse Architecture, Infrastructure and Metadata. Data Design and Data Representation: Principles of Dimensional Modeling, Data Extraction, Transformation and Loading, Data Quality, Online Analytical Processing (OLAP)–OLAP and Multidimensional Data Analysis.		
Unit: II	Data Mining – Pre-processing:	Contact Hours:11
Steps in Data mining process, Data Mining Functionalities, Architecture of a Typical Data Mining Systems, Classification of Data Mining Systems, Knowledge Discovery in Databases (KDD), KDD Process, Data Preprocessing, Data Transformation, Data Compression and Dimension Reduction, Principal Component Analysis, Binning Methods.		
Unit: III	Data Mining Techniques	Contact Hours:12
Association Rule Mining, Classification and Prediction: Efficient and Scalable Frequent Itemset Mining Methods, Mining, Various Kinds of Association Rules, Association Rules, Market Basket Analysis, Apriori Algorithm, Tree Based Algorithms. Classification by Decision Tree Introduction, Bayesian Classification, Rule Based Classification, Classification by Back propagation, Support Vector Machines with Machine learning, Lazy Learners, Prediction Techniques, Regression Models.		
Unit: IV	Clustering & Introduction to Web Mining	Contact Hours: 10
Data Mining Algorithms: Clustering. Partitioned Algorithms, Hierarchical Algorithms, Density Based, Algorithms, Grid Based Algorithms, Web Content Mining, Web Structure Mining, Web Usage Mining, Spatial Mining, Multimedia Data Mining, Text Mining.		



List of Experiment

Unit 1: Data Warehousing – Introduction and Design

Contact Hours: 8

1. Creating a Data Warehouse by defining Dimension and Fact Tables and demonstration of OLAP operations.
2. Implementation of Star Data Warehouse Schema.
3. Implementation of Fact Constellation Warehouse Schema.
4. Implementation of Snowflake Data Warehouse Schema.

Unit 2: Data Mining – Pre-processing

Contact Hours: 8

5. Demonstration of preprocessing operations on student.arff dataset.
6. Write a procedure for visualization of weather dataset/table.
7. Write a Python program to implement Bayes Theorem using probability analysis on colored ball datasets.
8. Write a program to solve the classical Water Jug Problem of Artificial Intelligence using Python.

Unit 3: Data Mining Techniques

Contact Hours: 8

9. Write a Python program to implement the Apriori algorithm for association rule mining.
10. Using KNN, construct a decision tree for employee/weather dataset and perform classification.
11. Create a Placement.arff dataset to identify students eligible for placements using KNN classification.
12. Write a Python program to solve the Travelling Salesman Problem using Artificial Intelligence techniques.

Unit 4: Web Data Mining

Contact Hours: 6

13. Write a Python program to implement K-Means clustering algorithm.
14. A case study of Business Intelligence in Government or Social sector applications.
15. Group Project: Development of an Intelligent Data Mining and Warehousing Framework for Decision Support.

TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)

Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	5 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 WEKA / Python Programming tasks	10	Based on the coding solutions submitted.
Real-World Case Study- Based Learning	Case study on mining a real dataset	10	Based on technical depth, data collection, analysis and findings.
Online Quizzes	5 Online Quiz	5	Based on quiz scores

OR

- **Online Courses:** NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).
 - Certificate & Report Mandatory Before End of Semester.
 - **If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)**

All Reports Must Be Pilgrims & AI Content Free as Per the Rule.



Textbooks:

1. J. Han and M. Kamber, “Data Mining Tools and Techniques”, Morgan Kaufmann Publishers.
2. M.H. Dunham, “Data Mining Introductory and Advanced Topics”, Pearson Education.

Reference Books:

1. Prabhu, “Data warehousing- concepts, Techniques, Products and Applications”, Prentice Hall of India.
2. Alex Berson and Stephen J. Smith, “Data Warehousing, Data Mining & OLAP”, Tata McGraw Hill Edition, Tenth Reprint.
3. Pang-Ning Tan, Michael Steinbach and Vipin Kumar, “Introduction to Data Mining”, Pearson Education.
4. Data Mining Techniques: For Marketing, Sales, and Customer Relationship Management: Gordon S. Linoff & Michael J. A. Berry, Wiley.

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**
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PANIPAT INSTITUTE OF ENGINEERING & TECHNOLOGY

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AN AUTONOMOUS INSTITUTE

Panipat Institute of Engineering and Technology An Autonomous Institute Affiliated to Kurukshetra University Kurukshetra

B.Tech Computer Science Engineering Semester-VI

Scheme of Studies & Examination (w.e.f session 2026)

Course No.	Course Title	Contact Hrs./Sem			TW / SL	Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P				Maximum Marks	Maximum Marks	Maximum Marks	
HS											
BBA-376A	Business Intelligence & Entrepreneurship	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
Professional Core Courses											
BT-CSE-302A /BT-CYS-453A / BT- IT-302A	Compiler Design	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-304A / BT-CYS-305A /BT-AIML-305A/BT-IT-305A/BT-AIDS-455A	Computer Networks	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-3***A	Open Elective-I	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-***A	Program Elective-I	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-374A	Computer Networks Lab	-	-	60	0	60	60/30=2	50	50	100	3 Hrs
BT-CSE-***-A	Elective-I Lab	-	-	60	0	60	60/30=2	50	50	100	3 Hrs
BT-CSE-376A	Project-1	-	-	60	0	60	60/30=2	50	50	100	3 Hrs
Total		225	0	180	225	630	21	350	450	800	
OE Elective-I		PEC Elective-I						PEC Elective LAB-I			
Advanced Computer Architecture: BT-CSE-352A		Fundamentals of Data Science: BT-CSE-358A						Fundamentals of Data Science Lab: BT-CSE-378A			



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Distributed Systems: BT-CSE-354A/BT-AIML-355A/BT-IT-362A	Unix & Linux Programming: BT-CSE-360A	Unix & Linux Programming Lab: BT-CSE-370A
Information Security & Cryptography: BT-CSE-356A	Fundamentals of Game Development using Blender BT-CSE-362A	Fundamentals of Game Development using Blender Lab: BT-CSE-372A

For Honours Basket 3 & Exit Option after 6th Semester Computer Science Engineering

Honours Basket 3

BT-CSE-364A / BT-IT-457A	Fundamentals of Deep Learning	45	-	30	45	120	120/30=4	40	60	100	
BT-CSE-366A	Data Engineering	45	-	30	45	120	120/30=4	40	60	100	

6th Semester Exit Course

Course No.	CourseTitle	Contact Hrs./Sem			TW/SL	Hours/Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P				Maximum Marks	Maximum Marks	Maximum Marks	
BT-CSE-401A	Neural Networks & Deep Learning/NPTL/ Project	45	-	-	45	90	90/30=3	40	60	100	3 Hrs
BT-CSE-473A	Summer Internship -III	-		-			3	50	50	100	3 Hrs
		45	0	0		90	6	90	110	200	3 Hrs
Grant Total (with Exit Option Without Honours)		525	0	270	480	1320	49	830	1070	1900	

Note: 06 weeks Summer internship-III is mandatory at the end of 6th semester (before the commencement of 7th semester).

Students Exiting the program after Securing 137.0 credit will be awarded U.G Degree of computer Science Engineering.

Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated honours Course Basket 3.



Course Code	Course Title	L	T	P	SL	Credits
BBA-376A	Business Intelligence and Entrepreneurship	45	0	0	45	90/30=3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Understand who entrepreneurs are and what competencies are needed to become an entrepreneur.
CO2	Understand insights into management, opportunity search, identification of a product, market feasibility studies, project finalization etc. required for small business enterprises
CO3	Write a report and do an oral presentation on topics such as product identification, business ideas, export marketing etc.
CO4	Understand the different financial and other assistance available for small industrial units.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	2	1	2	1	2
CO2	2	3	2	2	1	2	3	-
CO3	2	2	3	3	2	1	2	2
CO4	2	2	3	3	2	2	3	-
Avg	2.25	2.25	2.5	2.5	1.5	1.75	2.25	1

Course Outline

Unit-1: Introduction of Entrepreneurship	Contact Hours: 12
Concept and Definitions; Entrepreneurship and Economic Development; Classification and Types of Entrepreneurs; Entrepreneurial Competencies; Factors Affecting Entrepreneurial Growth – Economic & Non-Economic Factors; EDP Programmes; Entrepreneurial Training; Traits/Qualities of an Entrepreneur; Manager vs. Entrepreneur; Entrepreneurial Challenges.	
Unit-2: Opportunity Identification and Product Selection	Contact Hours: 11
Entrepreneurial Opportunity Search and Identification; Criteria to Select a Product; Conducting Feasibility Studies; Sources of Business Ideas; Marketing Plan: Conducting Marketing Research, Industry Analysis, Competitor Analysis, Market Segmentation and Positioning, Building a Marketing Plan, Marketing Mix, Launching a New Product: Methods of Project Appraisal, Project Report Preparation, Specimen of Project Report; Project Planning and Scheduling using Networking Techniques of PERT / CPM	
Unit-3: Small Enterprises and Enterprise Launching Formalities	Contact Hours: 10
Definition of Small Scale; Rationale; Objectives; Scope; SSI; Registration; NOC from Pollution Board; Machinery and Equipment Selection; Role of SSI in Economic Development of India; Major Problems Faced by SSI/MSMEs; Definition and Significance in Indian Economy; MSME Schemes; Challenges and Difficulties in Availing MSME Schemes.	
Unit-4: Role of Support Institutions and Management of Small Business	Contact Hours: 12
DIC, SIDO, SIDBI; Small Industries Development Corporation (SIDC); SISI; NSIC; NISBUD; State	



Financial Corporation (SFC); Venture Capital – Concept, Venture Capital Financing Schemes offered by various financial institutions in India; Entrepreneurship and Intellectual Property Rights (IPR) and their Importance (Patent, Copyright, Trademarks).

Case Studies: At least one in the whole course

TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)

Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	05 Assignments	20	Based on the assignments submitted.
Real-World Case Study-Based Learning	Case study on a successful tech startup / BI deployment	10	Based on technical depth, data collection, analysis and findings.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	10	Based on poster / chart quality and presentation skills and Summary Report
Online Quizzes	05 Online Quiz based on related subject	5	Based on quiz scores

OR

- **Online Courses:** NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).
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 - **If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)**

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Textbooks

1. Khanka S.S., Entrepreneurial Development, S.Chand & Co. Ltd., New Delhi, 2001.
2. Sangeeta Sharma, Entrepreneurship Development, PHI Learning Pvt. Ltd., 2016.
3. Desai, Vasant, “**Entrepreneurship Development**”, Himalaya Publishing House, Mumbai, 2019.

Reference Books:

1. Barringer, B., Entrepreneurship: Successfully Launching New Ventures, 3rd Edition, Pearson, 2011.
2. Bessant, J., and Tidd, J., Innovation and Entrepreneurship, 2nd Edition, John Wiley & Sons, 2011.
3. Desai, V., Small Scale Industries and Entrepreneurship, Himalaya Publishing House, 2011.
4. Donald, F.K., Entrepreneurship- Theory, Process and Practice, 9th Edition, Cengage Learning, 2014.

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Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-302A / BT-CYS-453A / BT- IT-302A	Compiler Design	45	0	0	45	90/30=3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Understand the phases of language processing and design lexical analyzers using regular expressions and finite automata.
CO2	Analyse various parsing techniques for context free grammar to construct syntax trees and handle ambiguity in programming languages.
CO3	Design semantic analyzers and generate intermediate code using attribute grammar, symbol tables and syntax directed translation schemes.
CO4	Apply code optimization techniques and manage run-time environments for efficient program execution and error recovery.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	3	–	–	–	1	–	1	2	2
CO2	3	3	3	2	3	1	–	–	2	1	2	3	2
CO3	3	3	3	2	3	1	–	1	2	1	2	3	3
CO4	3	3	3	2	3	1	2	1	3	1	3	3	3
Avg.	3.00	2.75	2.75	1.75	3.00	0.75	0.50	0.50	2.00	0.75	2.00	2.75	2.50

Course Outline

Unit-1: Introduction to Compiler Design and Lexical Analysis	Contact Hours: 12
Introduction to Language Processing System, Compiling Analysis of the Source Program, Phases of a Compiler, Compiler Construction Tools. Lexical Analysis: Role of Lexical Analyzer, Specification of Tokens, Recognition of Tokens, Introduction to Finite Automata and Regular Expression, Conversion of Regular Expression to NFA	
Unit-2: Syntax Analysis and Parsing Techniques	Contact Hours: 10
Syntax Analysis: Role of the Parser, Abstract Syntax Trees, Ambiguity in Context-Free Grammars, Types of Parsing: Top-Down Parsing, Recursive Descent Parsing, Back Tracking, Predictive Parsing, Bottom Up Parsing, Shift reduce Parsing, Operator Precedence Parsing, LR Parsers, SLR Parser, Canonical LR Parser, LALR Parser.	
Unit-3: Semantic Analysis and Intermediate Code Generation	Contact Hours: 11
Semantic Analysis: Semantic Errors, Attribute Grammar, Synthesized attributes, Static Allocation, Stack Allocation, Heap Allocation, Activation Trees, Symbol Table, Intermediate Code Generation and Code Intermediate languages, Declarations, Assignment Statements, Boolean Expressions, Case Statements, DAG representation of Basic Blocks, A simple Code generator from DAG, Issues in the Design of Code Generator.	



Unit-4: Code Optimization and Run-Time Environments			Contact Hours: 12
Code Optimization and Run Time Environments, Principal Sources of Optimization, Machine dependent & Machine independent Optimization, Optimization of Basic Blocks, Loop Optimization, Peephole Optimization, Introduction to Global Data Flow Analysis, Storage Organization, Static Storage Management, Heap Storage management, Parameter Passing. Error Recovery, Panic mode, Statement mode, Global correction.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	05 Assignments	20	Based on the assignments submitted.
Complex Problem Solving	2 Complex Problems Related to Subject	10	Based on the depth and quality of the solution submitted.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	10	Based on poster / chart quality and presentation skills and Summary Report
Online Quizzes	05 Online Quiz based on related subject	5	Based on quiz scores
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's). - Certificate & Report Mandatory Before End of Semester. If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
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Textbooks:

1. Alfred Aho, Ravi Sethi, Jeffrey D Ullman, "Compilers Principles, Techniques and Tools", Pearson Education Asia, 2018.
2. Allen I. Holub "Compiler Design in C", Prentice Hall of India, 2003.
3. C. N. Fischer and R. J. LeBlanc, "Crafting a compiler with C", Benjamin Cummings, 2003.
4. V Raghavan, "Principles of Compiler Design", Second Edition, Tata McGraw-Hill, 2018.
5. Henk Alblas and Albert Nymeyer, "Practice and Principles of Compiler Building with C", PHI, 2001.
6. Kenneth C. Loudon, "Compiler Construction: Principles and Practice", Thompson Learning, 2003

Reference Books:

1. Alfred V. Aho & Jeffrey D. Ullman, "Principles of Compiler Design" (The Green Book), Addison Wesley, 1977.
2. Dick Grune, Kees van Reeuwijk, Henri E. Bal, Cerial J.H. Jacobs, Koen G. Langendoen, "Modern Compiler Design", Springer, 2012.
3. Reinhard Wilhelm & Helmut Seidl, "Compiler Design: Theory, Tools and Implementation",



Springer, 2010.

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Course Code					Course Title					L	T	P	SL	Credits	
BT- CSE-304A / BT-CYS-305 A / BT-AIML-305A / BT-IT-305A / BT- AIDS-455A					Computer Networks					45	0	0	45	90/30=3	
										CIE		SEE		Total	
										40		60		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Understand fundamentals, architecture, and transmission techniques of computer networks.														
CO2	Analyse data link layer functions, error control, and medium access protocols.														
CO3	Apply routing and addressing techniques using various network layer protocols.														
CO4	Analyze transport and application layer protocols and apply security techniques such as cryptography and authentication.														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	3	2	2	2	2	–	–	–	–	1	2	3	2		
CO2	3	3	2	2	2	–	–	–	–	1	2	3	3		
CO3	3	3	3	2	2	–	–	–	1	1	2	3	3		
CO4	3	3	3	2	2	–	–	–	1	1	2	3	3		
Avg.	3.00	2.75	2.50	2.00	2.00	–	–	–	0.50	1.00	2.00	3.00	2.75		

Course Outline

Unit-1: Fundamentals of Computer Networks and Physical Layer	Contact Hours: 12
Introduction: Introduction to Computer Networks, Data Communication System and its components, Data Flow, Computer network and its goals, Types of computer networks: LAN, MAN, WAN, Wireless and Wired networks, broadcast and point-to-point networks, Network topologies, protocols, ISO-OSI reference model, TCP/IP architecture. Physical Layer: Bandwidth, Transmission Impairments: Attenuation, Distortion, Noise, Multiplexing: Frequency Division, Time Division, Wavelength Division, Introduction to Transmission Media: Twisted pair, Coaxial cable, Fiber optics, Wireless transmission (radio, microwave, infrared), Switching: Circuit Switching, Message Switching, Packet Switching & comparisons, ATM	
Unit-2: Data Link Layer and Medium Access Techniques	Contact Hours: 11
Data link layer: Error Control, Types of errors, framing (character and bit stuffing), error detection & correction methods; Flow control Protocols: Stop & wait, ARQ, sliding window protocols: Go-Back- N, Selective repeat. Medium access sub layer: Point to point protocol, HDLC, Random Access Protocols (ALOHA, Slotted ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Reservation, Polling, Token Passing), Channelization techniques (FDMA, TDMA, CDMA), Devices-repeaters, hubs, switches, Bridges, Router, Gateway.	



Unit-3: Network Layer Protocols and Routing	Contact Hours: 10		
Network layer: Addressing: Internet address, subnetting; Routing techniques, static vs. dynamic routing, routing table, Routing algorithms: distance vector routing, link state routing, Routing protocols: RIP, OSPF, BGP, Protocols: ARP, RARP, IPV4, IPV6, ICMP, IGMP, DHCP.			
Unit-4: Transport Layer, Application Protocols and Network Security	Contact Hours: 12		
Transport layer: Process to process delivery; UDP; TCP, RPC, Congestion control algorithm: Leaky bucket algorithm, Token bucket algorithm, choke packets; Quality of service: techniques to improve QoS. Application layer: DNS; SMTP, SNMP, FTP, HTTP & WWW. Security: Cryptography, user authentication, security protocols in internet, public key encryption algorithm, digital signatures.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing (numerical / problem-based)	05 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 socket-programming / Cisco Packet Tracer simulations	10	Based on the coding solutions submitted.
Real-World Case Study- Based Learning	Case study on network design / security incident	10	Based on technical depth, data collection, analysis and findings.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.
OR			
• Online Courses: NPTEL/Coursera/MooC’s/Swayam (As Per Available Hours/Credit’s). • Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Behrouz A. Forouzan, "Data communication and Networking", Tata McGraw Hill, Fourth Edition, 2011.
2. Computer Networks, 4th Edition, Pearson Education by Andrew S. Tanenbaum.

Reference Books:

1. Larry L. Peterson, Peter S. Davie, "Computer Networks", Elsevier, Fifth Edition, 2012.
2. William Stallings, "Data and Computer Communication", Eighth Edition, Pearson Education, 2007.
3. James F. Kurose, Keith W. Ross, "Computer Networking: A Top-Down Approach



Featuring the Internet”, Pearson Education, 2005.

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**
2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



Course Code				Course Title						L	T	P	SL	Credits	
BT-CSE-374A				Computer Networks Lab						0	0	60	0	60/30=2	
										CIE		SEE		Total	
										50		50		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Understand the fundamental networking components, topologies, and device interconnections using Packet Tracer.														
CO2	Design and configure IP addressing schemes, subnetting, and basic to advanced routing protocols (Static, RIP) to establish efficient network communication.														
CO3	Implement network services such as DHCP, DNS, Web, Email, FTP, and remote device access (SSH/Telnet) to enable end-to-end connectivity and service delivery.														
CO4	Analyze and troubleshoot network operations using Packet Tracer simulation tools and integrate emerging technologies like IoT into network environments.														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	3	2	2	2	2	–	–	–	–	1	2	3	2		
CO2	3	3	3	2	3	–	–	–	–	1	2	3	3		
CO3	3	3	2	2	3	–	–	–	1	2	2	3	3		
CO4	3	3	3	2	3	–	–	–	1	2	2	3	3		
Avg.	3.00	2.75	2.50	2.00	2.75	–	–	–	0.50	1.50	2.00	3.00	2.75		

List of Experiments:

1. Identify and understand routers, switches, hubs, bridges, gateways, and PCs in Packet Tracer.
2. Create and simulate Star, Bus, Ring, and Mesh topologies.
3. Connect two PCs using a crossover cable and Create LAN network using switch.
4. Design and assign IP addresses for a given network using subnetting.
5. Configure ARP and RARP protocols on Packet Tracer.
6. Implement static routing between two or more networks.
7. Configure RIP (Routing Information Protocol) on multiple routers.
8. Create a WAN network using multiple Routers.
9. Configure router or dedicated DHCP server for dynamic IP allocation.
10. Configure a DNS server for domain name resolution.
11. Configure and simulate web & email services.



12. Use Packet Tracer simulation mode to capture and analyze packets.
13. Create a simple IoT setup (smart lights, sensors) connected via a router.
14. Configure SSH and Telnet access on routers and switches.
15. Configure an FTP server and clients for file transfer.

Textbooks:

1. Behrouz A. Forouzan, “Data Communications and Networking”, 5th Edition, McGraw Hill Education, 2017.
2. James F. Kurose and Keith W. Ross, “Computer Networking: A Top-Down Approach”, 7th Edition, Pearson Education, 2016.
3. Cisco Networking Academy, “Routing and Switching Essentials”, Cisco Press, Latest Packet Tracer–Aligned Edition, 2020.

Reference Books:

1. Wendell Odom, “CCNA 200-301 Official Cert Guide, Vol. 1 & 2”, 1st Edition, Cisco Press, 2020.
2. Andrew S. Tanenbaum and David Wetherall, “Computer Networks”, 5th Edition, Pearson Education, 2011.
3. Behrouz A. Forouzan, “TCP/IP Protocol Suite”, 4th Edition, McGraw Hill Education, 2010.
4. Gordon Davies, “Networking Fundamentals”, 2nd Edition, Pearson Education, 2019.
5. Cisco Networking Academy, “Packet Tracer Network Simulator Guide”, Cisco Press, Latest Edition, 2021.
6. Cisco Networking Academy, “Introduction to IoT”, Online Curriculum, 2022.

NOTE: A student must perform at least Fifteen experiments. Twelve experiments should be performed from the above list. The remaining Three experiments may either be performed from the above list or as per the scope of the syllabus.



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(6th Sem)

Open

Elective - 1



Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-352A	Advanced Computer Architecture	45	0	0	45	90/30=3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Analyse computational models, pipeline structures, ILP concepts, and fundamental features of the RISC-V architecture.
CO2	Analyse superscalar execution techniques, branch-handling mechanisms, and domain-specific accelerators like GPUs/TPUs.
CO3	Analyse parallel architectures, interconnection networks, and memory/coherence mechanisms in scalable systems.
CO4	Analyse GPU and WSC architecture and apply basic CUDA/OpenCL concepts.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	2	—	—	—	—	1	1	3	2
CO2	3	3	2	2	2	—	—	—	—	1	1	3	3
CO3	3	3	2	2	3	—	—	—	—	1	1	3	3
CO4	3	3	2	3	3	—	—	—	—	1	1	3	3
Avg.	3.00	3.00	2.00	2.25	2.50	—	—	—	—	1.00	1.00	3.00	2.75

Course Outline

Unit-1: Pipelined & Parallel Computer Architectures	Contact Hours: 12
Basic computational models and the evolution of computer architecture. The concept of computer architecture as a multilevel hierarchical framework. Types and levels of parallelism. The principle and general structure of pipelines, performance measures. Instruction Level Parallel (ILP) processors and dependencies between instructions. Pipelined processing of integer, Boolean, load, and store instructions. RISC-V Architecture: Introduction to the open-source instruction set architecture and its key features.	
Unit-2: Superscalar & Domain-Specific Processor Architectures	Contact Hours: 12
The emergence of superscalar processors. Tasks of superscalar processing: parallel decoding, instruction issue, shelving, register renaming, parallel execution, and preserving sequential consistency. A comparison of VLIW and superscalar processors. The branch problem and approaches to branch handling include delayed branching, branch detection and prediction schemes, branch penalties, and guarded execution. Domain-Specific Architectures (DSAs) and Accelerators: Introduction to hardware accelerators for specific workloads, such as GPUs and TPUs.	
Unit-3: Parallel & Distributed Computer Architectures	Contact Hours: 11



Classification of parallel architecture. Concepts of distributed and shared memory MIMD architectures (UMA, NUMA, CC-NUMA, and COMA models). Problems with scalable computers. Static interconnection networks (linear array, ring, mesh, torus, hypercubes, fat tree) and dynamic interconnection networks (single shared buses, crossbar, and multistage networks). Advanced processor technology, memory hierarchy technology, and virtual memory. Cache coherence problems and hardware-based protocols (snoopy cache and directory schemes). Software-based protocols.

Unit-4: GPU and Warehouse-Scale Computing Architectures

Contact Hours: 10

GPU Computing: Architecture of GPUs and their use for general-purpose computing (GPGPU). Introduction to programming models like CUDA and OpenCL. Warehouse-Scale Computers (WSC): Architecture, Workloads, Physical Infrastructure; Basics of Cloud Computing Architectures, Enterprise Memory Subsystems: Fault tolerance, memory mirroring, and scrubbing

TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)

Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Self-Learning Online Course	Video Based on Advanced / Parallel Computer Architecture	10	Based On Summary Report
Real-World Case Study-Based Learning	Case study on GPU / multicore architectures	10	Based on technical depth, data collection, analysis and findings.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.

OR

Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).

- Certificate & Report Mandatory Before End of Semester.
- **If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)**

All Reports Must Be Pilgrims & AI Content Free as Per the Rule.

Textbooks

1. John L. Hennessy and David A. Patterson, Computer Architecture: A Quantitative Approach, Morgan Kaufmann Publishers (Elsevier), 6th Edition, 2019.
2. Dezső Sima, Terence Fountain, and Peter Kacsuk, Advanced Computer Architectures: A Design Space Approach, Pearson Education, 1st Edition, 1997.
3. Michael J. Quinn, Parallel Computing: Theory and Practice, McGraw-Hill Education, 2nd Edition, 1994.
4. Thomas G. Lewis and Hesham El-Rewini, Introduction to Parallel Computing, Prentice Hall, 2nd Edition, 1992.
5. Nicholas Carter, Computer Architecture, McGraw-Hill Education, Schaum's Outline Series Edition, 2002.

Reference Books

1. Kai Hwang, Advanced Computer Architecture: Parallelism, Scalability, Programmability, Tata



McGraw-Hill Education, 1st Edition, 1993.

2. William Stallings, Computer Organization and Architecture: Designing for Performance, Pearson Education, 11th Edition, 2019.
3. David E. Culler, Jaswinder Pal Singh, and Anoop Gupta, Parallel Computer Architecture: A Hardware/Software Approach, Morgan Kaufmann Publishers (Elsevier), 1st Edition, 1999.

E-Resource:

- NPTEL

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Course Code				Course Title						L	T	P	SL	Credits	
BT-CSE-354A BT-AIML-355A BT-IT-362A				Distributed Systems						45	0	0	45	90/30=3	
										CIE		SEE		Total	
										40		60		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Understand Fundamental concepts and synchronization of concurrent processes.														
CO2	Analyse methods to maintain consistency and resolve conflicts among processes.														
CO3	Apply strategies to ensure system reliability and recover from failures.														
CO4	Evaluate consensus algorithms and adaptive failure detectors.														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	3	3	2	2	2	1	–	–	1	1	–	2	2		
CO2	2	3	2	2	2	1	–	–	1	1	–	3	3		
CO3	2	2	3	3	2	2	1	2	–	1	1	3	3		
CO4	3	3	2	3	2	1	–	1	1	1	1	3	3		
Avg.	2.50	2.75	2.25	2.50	2.00	1.25	0.25	0.75	0.75	1.00	0.50	2.75	2.75		

Course Outline

Unit: 1 Distributed Basics and Synchronization	Contact Hours: 12
Distributed Systems: Introduction, Relation to parallel multiprocessor/multicomputer systems, Message-passing systems versus shared memory systems, Synchronous versus asynchronous executions, Design issues and challenges Distributed mutual exclusion algorithms: Lamport's algorithm, Ricart-Agrawala algorithm, Singhal's dynamic information-structure algorithm, Lodha and Kshemkalyani's fair mutual exclusion algorithm, Quorum-based mutual exclusion algorithms, Maekawa's algorithm, Agarwal-El Abbadi quorum-based algorithm, Token-based algorithms, Suzuki-Kasami's broadcast algorithm, Raymond's tree-based algorithm	
Unit: 2 Distributed Shared Memory and Deadlock Handling	Contact Hours:12
Deadlock detection: Introduction, Models of deadlocks, Knapp's classification of distributed deadlock detection algorithms, Mitchell and Merritt's algorithm for the single-resource model, Chandy-Misra-Haas algorithm for the AND model, Chandy-Misra-Haas algorithm for the OR model, Kshemkalyani-Singhal algorithm for the P-out-of-Q model Distributed shared memory Abstraction and advantages, Memory consistency models, Shared memory mutual exclusion, Wait-freedom, register hierarchy and wait-free simulations, Wait-free atomic snapshots of shared objects	



Unit: 3 Checkpointing and rollback recovery	Contact Hours:10		
Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Koo–Toueg coordinated checkpointing algorithm, Juang–Venkatesan algorithm for asynchronous checkpointing and recovery, Manivannan–Singhal quasi-synchronous checkpointing algorithm, Peterson–Kearns algorithm based on vector time, Helary–Mostefaoui–Netzer–Raynal communication-induced protocol			
Unit: 4 Failure Detectors, Consensus and Agreement Algorithms	Contact Hours:11		
Problem definition, Overview of results, Agreement in a failure-free system (synchronous or asynchronous), Agreement in (message-passing) synchronous systems with failures, Agreement in asynchronous message-passing systems with failures, Wait-free shared memory consensus in asynchronous systems Failure Detector: Introduction, Unreliable failure detectors, the consensus problem, atomic broadcast, A solution to atomic broadcast, the weakest failure detectors to solve fundamental agreement problems, An adaptive failure detection protocol			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Self-Learning Online Course	Video Based on Distributed System	10	Based On Summary Report
Real-World Case Study-Based Learning	Case study on a distributed system (Kafka / Hadoop)	10	Based on technical depth, data collection, analysis and findings.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.
OR			
Online Courses: NPTEL/Coursera/MooC’s/Swayam (As Per Available Hours/Credit’s).			
• Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Ajay D. Kshemkalyani and Mukesh Singhal, “Distributed Computing: Principles, Algorithms, and Systems”, Cambridge University Press, 1st Edition, 2008.
2. George Coulouris, Jean Dollimore, Tim Kindberg and Gordon Blair, “Distributed Systems: Concepts and Design”, Pearson Education, 5th Edition, 2011.
3. Nancy Lynch, “Distributed Algorithms”, Morgan Kaufmann Publishers, 1st Edition, 1996.

Reference Books:

1. Mukesh Singhal and Niranjana G. Shivaratri, “Advanced Concepts in Operating Systems:



Distributed, Database, and Multiprocessor Operating Systems”, McGraw Hill Education, 1st Edition, 1994.

2. Andrew S. Tanenbaum and Maarten Van Steen, “Distributed Systems: Principles and Paradigms”, Pearson Education, 2nd Edition, 2007.
3. Hagit Attiya and Jennifer Welch, “Distributed Computing: Fundamentals, Simulations, and Advanced Topics”, Wiley India, 2nd Edition, 2004.
4. Michel Raynal, “Distributed Algorithms for Message-Passing Systems”, Springer, 1st Edition, 2013.
5. Vijay K. Garg, “Elements of Distributed Computing”, Wiley India, 1st Edition, 2002.

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Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-356A	Information Security & Cryptography	45	0	0	45	3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Understand the fundamentals of information security, security models, cyber threats and access control mechanisms.
CO2	Apply mathematical foundations modular arithmetic, Euclidean algorithms and algebraic structures in cryptographic systems.
CO3	Analyze and implement symmetric and asymmetric cryptographic techniques for secure communication and data protection.
CO4	Evaluate network security mechanisms, malware protection techniques and modern cybersecurity solutions for real-world applications.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	–	–	1	1	–	2	–	–	–	2	1
CO2	3	3	–	1	2	–	–	–	–	–	–	2	2
CO3	3	3	2	2	3	1	–	1	–	–	–	3	2
CO4	3	3	3	2	3	2	1	2	1	1	1	3	3
Avg.	3.00	2.75	1.25	1.25	2.25	1.00	0.25	1.25	0.25	0.25	0.25	2.50	2.00

Course Outline

Unit-1	Information Security Fundamentals	Contact Hours: 12
Introduction to Information Security; Need and Goals of Information Security; Security Services and Mechanisms; CIA Triad; Threats, Vulnerabilities and Attacks; Types of Cyber Attacks; Components of Information Systems; Authentication, Authorization and Access Control; Security Policies and Risk Management Basics; OSI Security Architecture; Network Security Models; Network Access Security Model; Introduction to Cybersecurity Governance and Security Standards.		
Unit-2	Mathematical Foundations for Cryptography	Contact Hours: 11
Divisibility and Prime Numbers; Division Algorithm; Euclidean Algorithm and Extended Euclidean Algorithm; Fast Exponentiation Algorithm; Modular Arithmetic and Congruences; Chinese Remainder Theorem; Algebraic Structures; Groups, Rings and Fields; Galois Fields; Introduction to Elliptic Curve Cryptography (ECC).		
Unit-3	Modern Cryptography and Secure Communication	Contact Hours: 11



Introduction to Cryptography; Classical Encryption Techniques; Symmetric Key Cryptography; Block Cipher Principles; Data Encryption Standard (DES); Advanced Encryption Standard (AES); Block Cipher Modes of Operation; Public Key Cryptography; Trapdoor One-way Functions; RSA Algorithm; Diffie-Hellman Key Exchange; Elliptic Curve Cryptography Basics; Cryptographic Hash Functions; Message Authentication Codes (MAC); Digital Signatures; Public Key Infrastructure (PKI); Introduction to SSL/TLS and Secure Communication.

Unit-4 Network and Cyber Security

Contact Hours: 12

Network Security Concepts and Threats; Security Controls and Countermeasures; Firewalls and Intrusion Detection/Prevention Systems; Secure Programming Concepts; Protecting Programs and Data; Sniffing and Spoofing Attacks; Malware, Viruses, Worms and Ransomware; Honeypots and Security Monitoring; Web Security Basics, SQL Injection and Cross-Site Scripting (XSS); Cloud Security Fundamentals; SDN Security; IoT Security; Cybersecurity Trends and Ethical Issues.

TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)

Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 Python cryptography implementation tasks	10	Based on the coding solutions submitted.
Real-World Case Study-Based Learning	Case study on a cryptographic failure / breach	10	Based on technical depth, data collection, analysis and findings.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.

OR

Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).

- Certificate & Report Mandatory Before End of Semester.
- **If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)**

All Reports Must Be Pilgrims & AI Content Free as Per the Rule.

Textbooks:

1. Cryptography and Network Security: Principles and Practice, William Stallings, 7th Edition, Pearson Education.
2. Cryptography and Network Security, Behrouz A. Forouzan, Special Indian Edition, McGraw Hill Education.
3. Network Security Essentials: Applications and Standards, William Stallings, 6th Edition, Pearson Education.
4. Cyber Security, Nina Godbole and Sunit Belapure, 1st Edition, Wiley India.

References:

1. Applied Cryptography, Bruce Schneier, 2nd Edition, Wiley India.
2. Cryptography: Theory and Practice, Douglas R. Stinson and Maura Paterson, 4th Edition, CRC Press.



3. Guide to Computer Network Security, Joseph Migga Kizza, 5th Edition, Springer.
4. Computer Security Basics, Deborah Russell and G.T. Gangemi Sr., 2nd Edition, O'Reilly Publications.
5. Introduction to Modern Cryptography, Jonathan Katz and Yehuda Lindell, 2nd Edition, CRC Press.
6. Principles of Information Security, Michael E. Whitman and Herbert J. Mattord, 6th Edition, Cengage Learning.

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(6th Sem)

Program

Elective Course

(Elective-I)



Course Code				Course Title				L	T	P	SL	Credits	
BT-CSE-358A				Fundamentals of Data Science				45	0	0	45	90/30=3	
								CIE		SEE		Total	
								40		60		100	
Course Outcomes: After Completion of Course, Students will be able to													
CO1	Understand concepts of data, Data Science lifecycle, and Big Data characteristics.												
CO2	Apply data collection and preprocessing techniques using Big Data tools.												
CO3	Perform data handling and exploratory data analysis using NumPy and Pandas.												
CO4	Create effective data visualizations for interpretation and decision-making.												
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)													
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	–	–	–	1	–	–	–	1	–	1	1
CO2	2	3	2	2	3	–	–	–	–	1	1	2	1
CO3	3	3	2	3	3	–	–	–	–	1	1	2	2
CO4	2	3	2	2	2	–	–	–	–	3	1	2	2
Avg.	2.50	2.75	1.50	1.75	2.00	0.25	–	–	–	1.50	0.75	1.75	1.50

Course Outline

Unit-1: Data, Data Science & Big Data	Contact Hours: 11
Definitions: Data, Information, Digital Data, Classification of Digital Data (structured / unstructured / semi-structured), Data Science: History and evolution , Roles in Data Science (analyst, engineer, scientist), Data Science process/lifecycle: data collection, preprocessing, modeling, analysis, deployment/insights, Applications of Data Science (business, healthcare, finance, social media, etc.), Big Data: definition, characteristics (volume, velocity, variety, veracity).	
Unit-2: Data Collection, Preprocessing & Big Data Technologies	Contact Hours: 11
Overview of data collection methods (manual, automated, APIs, web scraping, sensors, logs), Data preprocessing: cleaning, handling missing data, outliers; encoding categorical variables; normalization/scaling, Introduction to Big Data tools: Why Hadoop?, Basics of Hadoop architecture (HDFS, MapReduce / YARN) and major ecosystem components , NoSQL Databases: definition, need, types (Key-Value, Document, Column-family, Graph), use-cases, SQL vs NoSQL tradeoffs, When to use Big Data / NoSQL vs conventional RDBMS — industry considerations.	
Unit-3: Data Handling; Exploratory Data Analysis (EDA)	Contact Hours: 11
Introduction to NumPy: creating arrays, indexing/slicing, vectorized operations, Introduction to Pandas: Series, DataFrame, loading data, data manipulation (selecting, filtering, adding/removing columns), merging/joining, grouping & aggregation, Exploratory Data Analysis (EDA): descriptive statistics (mean, median, variance, std deviation), skewness, kurtosis, correlation analysis, ANOVA.	



Unit-4: Data Visualization			Contact Hours: 10
Importance of visualization in Data Science, Role of visualization in decision-making and reporting, Principles of effective visualizations: clarity, simplicity, accuracy, storytelling, Matplotlib: Basic plots: line, bar, scatter, histogram, Customizing plots: titles, labels, legends, colors, markers, Figure and axes objects, subplots, styling, Advanced Data Representation Techniques: Aggregation and grouped visualizations, Time series visualization basics, Pivot tables and summarized views.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 Python Data Science Task	10	Based on the coding solutions submitted.
Real-World Case Study-Based Learning	End to End Case Study on real dataset	10	Based on technical depth, data collection, analysis and findings.
Online Technical Quiz	Online Quix Based on Subject Syllabus	5	Based on quiz scores.
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).			
• Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Wes McKinney, Python for Data Analysis, O'Reilly Media, 2nd Edition.
2. Joel Grus, Data Science from Scratch: First Principles with Python, O'Reilly Media, 2nd Edition.
3. Samuel Burns / Addisu Taye Tesfaye, Fundamentals of Data Science, Latest Edition.

Reference books:

1. Peter Bruce, Andrew Bruce, Peter Gedeck, Practical Statistics for Data Scientists, O'Reilly Media, 2nd Edition.
2. Jake VanderPlas, Python Data Science Handbook, O'Reilly Media, 1st Edition.
3. Nathan Marz, James Warren, Big Data: Principles and Best Practices of Scalable Real-Time Data Systems, Manning Publications.
4. Tom White, Hadoop: The Definitive Guide, O'Reilly Media, 4th Edition.

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2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



Course Code				Course Title						L	T	P	SL	Credits	
BT-CSE-378A				Fundamentals of Data Science Lab:						0	0	60	0	60/30=2	
										CIE		SEE		Total	
										40		60		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1		Understand concepts of data, Data Science lifecycle, and Big Data technologies.													
CO2		Apply data collection, preprocessing, and Big Data tools for data management.													
CO3		Perform data handling and exploratory data analysis using NumPy and Pandas.													
CO4		Create and interpret data visualizations for effective decision-making.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	3	2	–	–	–	1	–	–	–	1	–	2	1		
CO2	2	3	2	2	3	–	–	–	–	1	1	2	2		
CO3	3	3	2	3	3	–	–	–	–	1	1	3	2		
CO4	2	3	2	2	2	–	–	–	–	3	1	2	2		
Avg.	2.50	2.75	1.50	1.75	2.00	0.25	–	–	–	1.50	0.75	2.25	1.75		

List of Experiments

1. Study and classification of structured, semi-structured, and unstructured data.
2. Analysis of a real-world dataset using the Data Science lifecycle.
3. Comparative study of Big Data characteristics (Volume, Velocity, Variety, Veracity).
4. Data collection using APIs and web scraping techniques.
5. Data cleaning and handling missing values in a dataset.
6. Detection and treatment of outliers using statistical methods.
7. Data transformation using encoding, normalization, and scaling techniques.
8. Installation and exploration of Hadoop ecosystem components and HDFS operations.
9. Comparative study of SQL and NoSQL databases using sample datasets.
10. Implementation of NumPy arrays, indexing, slicing, and vectorized operations.
11. Data manipulation using Pandas DataFrame and Series.
12. Data merging, grouping, aggregation, and filtering using Pandas.
13. Exploratory Data Analysis using descriptive statistics, correlation analysis, skewness, and kurtosis.



14. Creation of basic visualizations using Matplotlib (line, bar, scatter, histogram).
15. Development of advanced visualizations using subplots, grouped charts, pivot tables, and time-series data.

Textbooks

1. Wes McKinney, Python for Data Analysis, O'Reilly Media, 2nd Edition.
2. Joel Grus, Data Science from Scratch: First Principles with Python, O'Reilly Media, 2nd Edition.
3. Samuel Burns and Addisu Taye Tesfaye, Fundamentals of Data Science, Springer, Latest Edition.

Reference Books

1. Peter Bruce, Andrew Bruce, and Peter Gedeck, Practical Statistics for Data Scientists, O'Reilly Media, 2nd Edition.
2. Jake VanderPlas, Python Data Science Handbook, O'Reilly Media, 1st Edition.
3. Nathan Marz and James Warren, Big Data: Principles and Best Practices of Scalable Real-Time Data Systems, Manning Publications, 1st Edition.
4. Tom White, Hadoop: The Definitive Guide, O'Reilly Media, 4th Edition

NOTE: A student must perform at least Fifteen experiments. Twelve experiments should be performed from the above list. The remaining Three experiments may either be performed from the above list or as per the scope of the syllabus.



Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-360A	UNIX and Linux Programming	45	0	0	45	90/30=3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Apply Unix/Linux commands, file management techniques, and basic system administration operations effectively.
CO2	Develop and debug programs using Unix/Linux shell scripting, text-processing tools, and a C/C++ development environment, and automate tasks effectively
CO3	Apply process management, signal handling, job control, and interprocess communication techniques in Linux.
CO4	Apply system administration, performance monitoring, and security tools effectively.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	1	3	—	—	1	—	2	1	2	1
CO2	3	3	2	2	3	—	—	1	1	2	1	3	2
CO3	3	3	2	2	3	—	—	1	1	2	1	2	2
CO4	2	3	2	2	3	2	1	2	1	2	2	2	2
Avg.	2.75	2.75	1.75	1.75	3.00	0.50	0.25	1.25	0.75	2.00	1.25	2.25	1.75

Course Outline

Unit-1: Foundations & Basic Commands	Contact Hours: 12
History of Unix, Unix vs Linux, Features of Unix System, Architecture of Unix, Types of shells, Unix Booting procedure(startup and shutdown), User Accounts, Unix file system (types of files, symbolic links, pipes, sockets), Accessing File system (mounting/ unmounting), Structure of file system, Basic Unix commands, Advance Unix commands (File access permissions- chmod, chown, chgrp) , File compression techniques: zip, unzip, pack, unpack.	
Unit-2: Shell, Editors and Programming Environments	Contact Hours: 11
Shell Programming/ Scripting: writing scripts(bash), variables, loops, conditional, functions; Regular expression pattern (grep, egrep, fgrep), Editors: vi/vim, Stream editor (sed), AWK for processing structured text, Perl scripting basics for text processing. The C /C++ Environment: Compiling C/C++ programs, GCC/G++ Compiler options, debugging tools Makefiles, Static vs shared libraries, dynamic loading / linking.	



Unit-3: Process and Signal and IPC	Contact Hours: 12		
Processes in Linux: rc and init files; system startup, initialization scripts, Process concepts and lifecycle; Understanding process IDs, parent-child relationships, Job Control and Scheduling: Signal handlers overview, Job control commands: fg, bg jobs, Scheduled jobs: at, batch, cron, time. Signals in Unix/Linux: Concept and types of signals: standard, real-time, user-defined, Signal functions signal(), sigaction(),Signal-related system calls: kill(), raise(), alarm(), pause(), abort(), sleep() Interprocess Communication (IPC): Introduction: need for IPC,Types of IPC.			
Unit-4: Advanced System Administration & Security	Contact Hours: 10		
System performance monitoring & tuning (top, iostat, vmstat, sar, htop), Kernel configuration & tuning parameters (/proc, sysctl), Centralized log management & log rotation (rsyslog, journalctl). Networking tools: ping, telnet, ftp, route, Firewall, Intrusion detection basics (Snort, Fail2ban), Backup & recovery tools (tar, cpio, dd utility)			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 Shell-Scripting	10	Based on the coding solutions submitted.
Online Technical Quizzes	5 Linux & shell quizzes	10	Based on quiz scores.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.
OR			
Online Courses: NPTEL/Coursera/MooC’s/Swayam (As Per Available Hours/Credit’s).			
• Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Sumitabha Das, "Unix: Concepts and Applications", 4th Edition, McGraw Hill Education, 2017.
2. Neil Matthew and Richard Stones, "Beginning Linux Programming", 4th Edition, Wrox Press, 2008.
3. Richard Stevens, Stephen A. Rago, "Advanced Programming in the UNIX Environment (APUE)", 3rd Edition, Addison-Wesley, 2013.

Reference Books:

1. W. Richard Stevens, "UNIX Network Programming: Volume 1 – The Sockets Networking API", 3rd Edition, Pearson Education, 2004.



2. Brian W. Kernighan and Rob Pike, “The UNIX Programming Environment”, Prentice Hall, 1984.
3. Mark Sobell, “A Practical Guide to Linux Commands, Editors, and Shell Programming”, 4th Edition, Addison-Wesley, 2017.
4. Evi Nemeth, Garth Snyder, Trent R. Hein, Ben Whaley, “UNIX and Linux System Administration Handbook”, 5th Edition, Pearson Education, 2018.
5. Arnold Robbins, “Linux Programming by Example: The Fundamentals”, Prentice Hall, 2003.
6. Daniel Barrett, Richard Silverman, Robert Byrnes, “Linux Pocket Guide”, 3rd Edition, O’Reilly Media, 2016.

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**
2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



Course Code				Course Title					L	T	P	SL	Credits	
BT-CSE-370A				UNIX and Linux Programming Lab					0	0	60	0	60/30=2	
									CIE		SEE		Total	
									40		60		100	
Course Outcomes: After Completion of Course, Students will be able to														
CO1	Apply Unix/Linux commands to manage the file system, users, permissions, and basic administrative operations.													
CO2	Develop and execute shell scripts using variables, loops, conditionals, functions, and text-processing tools such as grep, sed, and awk.													
CO3	Analyze Linux process management by exploring process hierarchies, job control, scheduling mechanisms, and signal handling in C programs.													
CO4	Use system utilities and performance-monitoring tools to evaluate CPU, memory, and I/O usage and troubleshoot system behavior.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	
CO1	3	2	1	1	3	–	–	1	–	2	1	2	1	
CO2	3	3	2	2	3	–	–	1	1	2	1	3	2	
CO3	3	3	2	2	3	–	–	1	1	2	1	2	2	
CO4	2	3	2	2	3	2	1	2	1	2	2	2	2	
Avg.	2.75	2.75	1.75	1.75	3.00	0.50	0.25	1.25	0.75	2.00	1.25	2.25	1.75	

List of Experiment

1. Perform File System Operations for Directory Navigation, File Management, and File Type Identification.
2. Configure User Accounts, Groups, and File Access Permissions in Linux.
3. Explore Linux Shell Environment and Execute Basic System Administration Commands.
4. Perform File Compression, Archiving, and Extraction using Linux Utilities.
5. Develop Shell Scripts using Variables, Loops, Conditional Statements, and Functions.
6. Implement Text Processing and Pattern Matching using grep, sed, and awk Utilities.
7. Implement Program Compilation, Linking, and Debugging using GCC/G++ and GDB Tools.
8. Analyze Linux Process Creation, Process Hierarchies, and Initialization Mechanisms.
9. Implement Job Control and Task Scheduling using fg, bg, jobs, at, batch, and cron Commands.
10. Implement Signal Handling Mechanisms using signal() and sigaction() System Calls.
11. Perform System Performance Monitoring and Resource Utilization Analysis using top, htop, iostat, and vmstat Utilities.
12. Configure Network Connectivity and Troubleshoot Network Operations using ping, telnet, ftp,



and route Commands.

13. Implement Interprocess Communication using Pipes and FIFO Mechanisms.
14. Implement Interprocess Communication using Shared Memory and Message Queues.
15. Perform Backup, Recovery, and System Log Management using tar, dd, rsyslog, and journalctl Utilities.

Textbooks

4. Sumitabha Das, UNIX: Concepts and Applications, McGraw Hill Education, 4th Edition, 2017. ISBN: 978-1259006029.
5. Neil Matthew and Richard Stones, Beginning Linux Programming, Wrox Press, 4th Edition, 2008. ISBN: 978-0470147627.
6. W. Richard Stevens and Stephen A. Rago, Advanced Programming in the UNIX Environment, Addison-Wesley Professional, 3rd Edition, 2013. ISBN: 978-0321637734.

Reference Books

5. W. Richard Stevens, UNIX Network Programming, Volume 1: The Sockets Networking API, Addison-Wesley Professional, 3rd Edition, 2003. ISBN: 978-0131411555.
6. Brian W. Kernighan and Rob Pike, The UNIX Programming Environment, Prentice Hall, 1984. ISBN: 978-0139376818.
7. Mark G. Sobell, A Practical Guide to Linux Commands, Editors, and Shell Programming, Addison-Wesley Professional, 4th Edition, 2017. ISBN: 978-0134774602.
8. Evi Nemeth, Garth Snyder, Trent R. Hein, Ben Whaley, and Dan Mackin, UNIX and Linux System Administration Handbook, Pearson Education, 5th Edition, 2017. ISBN: 978-0134277554.
9. Arnold Robbins, Linux Programming by Example: The Fundamentals, Prentice Hall, 2004. ISBN: 978-0131429642.
10. Daniel J. Barrett, Richard E. Silverman, and Robert G. Byrnes, Linux Pocket Guide, O'Reilly Media, 3rd Edition, 2016. ISBN: 978-1491927571

NOTE: A student must perform at least Fifteen experiments. Twelve experiments should be performed from the above list. The remaining Three experiments may either be performed from the above list or as per the scope of the syllabus.



Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-362A	Fundamentals of Game Development using Blender	45	0	0	45	90/30=3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Explain the history, evolution, and development pipeline of computer games, and describe the significance of 3D assets in modern game design.
CO2	Demonstrate proficiency in Blender fundamentals, interface, editors, navigation tools, workspaces, and file handling standards.
CO3	Apply 3D modeling, texturing, lighting, shading, and animation techniques in Blender to create optimized game-ready assets.
CO4	Integrate, render, and optimize Blender assets into game engines (Unity/Unreal) to design and present a prototype interactive game scene.

Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	2	1	1	1	1	2	–	2	2
CO2	3	2	2	1	3	1	–	–	1	1	–	3	2
CO3	3	2	3	2	3	1	–	–	2	1	–	3	3
CO4	3	3	3	2	3	1	–	–	3	2	2	3	3
Avg.	3.00	2.25	2.50	1.50	2.75	1.00	0.25	0.25	1.75	1.50	0.50	2.75	2.50

Course Outline

Unit-1:	Game Development Overview & Blender Basics	Contact Hours: 10
History and evolution of computer games (2D → 3D → VR/AR), Game Development Pipeline: pre-production, production, post-production. Role of 3D assets in modern game design. Introduction to Blender: interface, editors, navigation tools. Workspaces: modeling, shading, animation. File handling: blend format, import/export standards (FBX, OBJ, GLTF).		
Unit-2:	3D Modeling and Texturing	Contact Hours: 11
3D modeling fundamentals in Blender, Mesh editing: Extrude, bevel, loop cut, knife tool. Modifiers: Mirror, subdivision, Boolean, array. Sculpting workflows in Blender. UV unwrapping: seams, islands, UVs Packing. Materials: PBR workflow (diffuse, metallic, roughness, normal maps). texture painting and procedural textures using shader nodes.		
Unit-3:	Lighting & Animation	Contact Hours: 12
Lighting in Blender: Types of lights: point, sun, spot, and area light, shadows, and rendering effects. Cameras: perspective vs orthographic, focal length, depth of field. Animation: keyframes, interpolation, timeline, graph editor. Basic animation techniques and Character Rigging: bones, armatures, weight painting, IK vs FK.		



Unit-4:	Rendering, Optimization & Game Engine Integration	Contact Hours: 12	
Rendering workflows: Eevee and Cycles engines, Texture baking and asset optimization, Polygon reduction and performance considerations, Exporting Blender assets to Unity and Unreal Engine, importing assets and testing compatibility, Designing a simple outdoor game environment, Mini- project: asset creation, integration, and prototype game scene.			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Video Based learning	Video Based on Blender	10	Based on Summary Report Submitted
Online Technical Quizzes	5 quizzes on modelling, texturing, rigging & animation (Blender)	5	Based on quiz scores.
Micro Project	Game Development	10	Based on Project Report
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).			
• Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Jason van Gumster, Blender for Dummies (4th Edition), Wiley, 2020.
2. Oliver Villar, Learning Blender: A Hands-On Guide to Creating 3D Animated Characters (2nd Edition), Addison-Wesley, 2019.

Reference Books:

1. John M. Blain, The Complete Guide to Blender Graphics: Computer Modeling and Animation (7th Edition), CRC Press, 2023.
2. Joseph Hocking, Unity in Action: Multiplatform Game Development in C# with Unity (2nd Edition), Manning, 2018.
3. Tracy Fullerton, Game Design Workshop: A Playcentric Approach to Creating Innovative Games (4th Edition), CRC Press, 2019.
4. Lee Zhi Eng, Building a Game with Unity and Blender, Packt, 2015.
5. Kumsal Obuz, Game Development with Blender and Godot, Packt, 2021.

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall



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be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. All questions shall carry equal marks.

2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



Course Code		Course Title								L	T	P	SL	Credits	
BT-CSE-372A		Fundamentals of Game Development using Blender Lab								0	0	60	0	60/30=2	
										CIE		SEE		Total	
										50		50		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Use Blender’s interface, tools, and primitives to model basic and intermediate 3D objects.														
CO2	Apply texturing, UV unwrapping, materials, lighting, and rendering techniques to create visually enhanced 3D scenes.														
CO3	Create animations, rig characters, and simulate physics-based effects using Blender’s animation and simulation tools.														
CO4	Export 3D assets and integrate them into game engines to design and present interactive game scenes.														
Course Outcomes (CO) to Program Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2		
CO1	2	2	3	1	3	–	–	–	1	2	1	3	1		
CO2	2	2	3	2	3	–	1	1	1	2	1	3	1		
CO3	2	3	3	2	3	–	–	1	2	2	1	3	2		
CO4	2	3	3	2	3	–	–	2	3	3	2	3	3		
Avg.	2.00	2.50	3.00	1.75	3.00	–	0.25	1.00	1.75	2.25	1.25	3.00	1.75		

List of Experiments:

1. Explore Blender Interface, Navigation, Editors, and Workspaces – Learn Blender UI and editors by navigating the 3D Viewport, Outliner, Timeline, and Properties Editor.
2. Create and Manipulate Basic Geometric Primitives – Practice creating cubes, spheres, cylinders, and cones, and apply translate, rotate, and scale transformations.
3. Design a Simple Indoor Scene Using Primitive Objects – Combine basic primitives to model a simple room environment.
4. Model a Game Prop Using Mesh Editing Techniques – Create a detailed game prop (e.g., weapon or tool) using extrusion, bevel, and inset operations.
5. Apply Blender Modifiers – Enhance 3D models using Mirror, Subdivision, and Boolean modifiers.
6. Perform UV Unwrapping and Apply Textures – Map 2D textures accurately onto 3D models using UV unwrapping.
7. Create Procedural Materials and Texture Painting – Apply procedural shaders and paint textures to 3D assets.
8. Add and Adjust Lighting in a 3D Scene – Place and modify point, spot, area, and sun lights to achieve desired visual effects.



9. Create a Keyframe Animation for a Moving Object – Animate a bouncing ball or moving car along a path using keyframes and interpolation.
10. Rig and Animate a Simple Character – Build a basic armature, assign weights, and animate character movements.
11. Apply Cloth and Particle Simulations – Simulate cloth movements and particle systems for effects like sparks or fountains.
12. Set Up Cameras and Render a Scene – Position cameras, adjust rendering settings, and generate images or video using Eevee/Cycles.
13. Export Assets and Import into Game Engine – Export Blender models in FBX/OBJ/GLTF formats and test them in Unity/Unreal Engine.
14. Develop a Mini Game Scene and Present – Integrate models, textures, lighting, and animations into a prototype game scene and present the final project.
15. Conduct a Team-Based Prototype Project Using Blender and a Game Engine.

Textbooks

1. John M. Blain, “The Complete Guide to Blender Graphics: Computer Modeling and Animation”, 7th Edition, CRC Press, 2023.
2. Ben Simonds, “Blender Master Class: A Hands-On Guide to Modeling, Sculpting, Materials, and Rendering”, No Starch Press, 2013.
3. Chris Totten, “Game Character Creation with Blender and Unity”, Wiley, 2012.

Reference Books:

1. Enrique Cabello et al., “Blender 3D by Example”, 2nd Edition, Packt Publishing, 2017.
2. Tony Mullen, “Introducing Character Animation with Blender”, 4th Edition, Wiley, 2020.
3. Richard Culver, “Blender Cycles: Materials and Textures Cookbook”, Packt Publishing, 2013.
4. Gordon Fisher, “Blender 3D Basics”, Packt Publishing, 2012.
5. Alan Thorn, “Mastering Unity: Build High-Quality Games”, Packt Publishing, 2015.

NOTE: A student must perform at least Fifteen experiments. Twelve experiments should be performed from the above list. The remaining Three experiments may either be performed from the above list or as per the scope of the syllabus.



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(6th Sem) Honours Course (Basket-3)



Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-364A / BT-IT-457A	Fundamentals of Deep Learning	45	0	30	45	90/30=3
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Understand the fundamentals of image formation, feature extraction, and visual matching techniques used in computer vision.
CO2	Apply deep learning concepts and analyze the evolution and behavior of CNN architectures.
CO3	Apply CNN-based methods for object recognition, detection, and segmentation.
CO4	Evaluate deep generative models and recent trends in computer vision.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	2	2	1	1	1	1	2	1	1	1
CO2	3	3	3	2	3	1	2	1	1	2	1	1	2
CO3	3	3	3	2	3	1	2	1	2	2	1	1	3
CO4	3	3	3	2	3	1	2	1	2	2	2	1	3
Avg.	3.00	2.75	2.75	2.00	2.75	1.00	1.75	1.00	1.50	2.00	1.25	1.00	2.25

Course Outline

Unit-1: Introduction	Contact Hours: 12
Introduction to Deep Learning, History and Evolution of Deep Learning, Features and Characteristics of Deep Learning, Deep Learning Techniques and Architectures, Deep Neural Networks and CNN Fundamentals, Optimization and Regularization Techniques, Transfer Learning and Data Augmentation, Evaluation Metrics for Deep Learning Models, Deep Learning Frameworks, Deep Learning Infrastructure and Computing Environment	
Unit-2: Deep Learning Architectures and Representation Learning	Contact Hours: 11
Evolution of Deep Learning Architectures: AlexNet, ZFNet, VGGNet, Inception Networks, ResNet, DenseNet and EfficientNet; Feature Extraction and Representation Learning; Deep Metric Learning: Siamese Networks, Contrastive Loss, Triplet Loss and Ranking Loss; Visualization and Understanding of Deep Networks using Feature Maps, Kernel Visualization, CAM and Grad-CAM.	
Unit-3: Deep Learning for Advanced Vision Tasks	Contact Hours: 12
Deep Learning-based Recognition and Verification; Object Detection using R-CNN, Fast R-CNN, Faster R-CNN, YOLO, SSD and RetinaNet; Semantic and Instance Segmentation using FCN, SegNet, U-Net and Mask R-CNN; Evaluation Metrics: Precision, Recall, IoU, mAP and Dice Coefficient; Real-world Applications of Deep Learning.	
Unit-4: Advanced Deep Learning and Generative AI	Contact Hours: 10
Deep Generative Models: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs) and Diffusion Models; Transformers and Attention Mechanism; Vision Transformers (ViT); Introduction to Large Language Models (LLMs), Foundation Models and Multimodal AI Systems; Self-supervised Learning; Zero-shot, One-shot and Few-shot Learning; Reinforcement Learning Concepts in Deep Learning; Recent Trends and Applications of Generative AI.	



List of Experiment

Unit 1: Foundations of Deep Learning

Contact Hours: 6

1. Install and configure deep learning frameworks such as TensorFlow, Keras and PyTorch.
2. Implementation of Deep Neural Network (DNN) for Classification Problems.
3. Develop an Image Classification using Convolutional Neural Network (CNN) Model.

Unit 2: Deep Learning Architectures and Representation Learning

Contact Hours: 8

4. Implementation and Performance Analysis of AlexNet Architecture.
5. Implementation of VGGNet, ResNet and DenseNet using Pre-trained Models.
6. Feature Extraction and Visualization of Feature Maps using CNN Models.
7. Implementation of Siamese Network using Contrastive Loss or Triplet Loss.

Unit 3: Deep Learning for Advanced Vision Tasks

Contact Hours: 8

8. Implement real-time object detection using deep learning-based models such as YOLO or SSD.
9. Implement object detection using Faster R-CNN framework.
10. Perform semantic image segmentation using U-Net architecture on sample image datasets.
11. Evaluate detection and segmentation models using IoU and mAP performance metrics.

Unit 4: Advanced Deep Learning and Generative AI

Contact Hours: 8

12. Implement Generative Adversarial Network (GAN) for image generation tasks.
13. Generate images using Variational Autoencoder (VAE) models.
14. Implement Vision Transformer (ViT) for image classification applications.
15. Perform experiments on Large Language Models (LLMs) using Hugging Face framework.

TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)

Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Video Based learning	Video Based on Deep Learning	10	Based on Summary Report Submitted
Online Technical Quizzes	5 TensorFlow / PyTorch model-building tasks	5	Based on quiz scores.
Micro Project	Deep Learning	10	Based on Project Report

OR

Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).

- Certificate & Report Mandatory Before End of Semester.
- **If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)**

All Reports Must Be Pilgrims & AI Content Free as Per the Rule.

Textbooks:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, "Deep Learning", MIT Press, 2017.
2. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 2nd Edition, O'Reilly, 2019.



3. Kevin P. Murphy "Machine Learning: A Probabilistic Perspective", The MIT Press, 2012.

Reference Books:

1. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017
2. Umberto Michelucci "Applied Deep Learning. A Case-based Approach to Understanding Deep Neural Networks" Apress, 2018.
3. Michael Bernico, *Practical Deep Learning for Cloud, Mobile, and Edge*, O'Reilly, 2020.
4. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, Prentice Hall of India, Third Edition 2014.

Instructions for Semester-End Examination

1. The semester-end examination shall consist of **nine questions** in total. **Question No. 1** shall be **compulsory** and will comprise **six short-answer questions** covering the entire syllabus. The remaining **eight questions** shall be distributed by setting **two questions from each of the four units** of the syllabus. Candidates shall be required to attempt **five questions** in total, including the compulsory **Question No. 1** and **one question from each of the four units**. **All questions shall carry equal marks.**
2. Candidates shall be permitted to use **only non-programmable scientific calculators** during the examination. The **sharing or exchange** of calculators or any other material is strictly prohibited. The use of **programmable calculators, mobile phones, smart watches, or any other electronic or communication devices** inside the examination hall is strictly prohibited.



Course Code	Course Title	L	T	P	SL	Credits
BT-CSE-366A	Data Engineering	45	0	45	30	120/30=4
		CIE		SEE		Total
		40		60		100

Course Outcomes: After Completion of Course, Students will be able to

CO1	Understand the various data collection and processing methods on different types of data e.g. text, video, audio,
CO2	Design databases, implement indexing techniques, and apply information retrieval models for efficient search and evaluation
CO3	Implement and analyze basic and advanced data structures and algorithms and evaluate their time and space complexity in real-world data processing applications.
CO4	Apply and analyze statistical and dimensionality reduction techniques, visualize and interpret data, and design meaningful data representations.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	2	3	1	–	–	1	1	1	3	2
CO2	3	3	3	2	3	1	–	–	1	1	1	3	3
CO3	3	3	3	2	3	1	–	–	2	1	1	3	3
CO4	3	3	3	2	3	1	–	–	2	1	1	3	3
Avg.	3.00	2.75	2.75	2.00	3.00	1.00	–	–	1.50	1.00	1.00	3.00	2.75

Course Outline

Unit-1: Data Collection and Preprocessing	Contact Hours: 10
Data Collection: Sources of data: Text, Video, Audio, Biological Data, Web scraping, APIs, Structured vs. unstructured data, Data acquisition tools and techniques, Sensor-based and IoT data collection Data Preprocessing: Handling Missing Data: Imputation techniques (mean, median, KNN, etc.), Noise Elimination and Outlier Detection, Feature Engineering and Feature Selection, Dimensionality Reduction: PCA, t-SNE, LDA, Normalization and Standardization techniques, Data Transformation (Log transformations, Encoding, etc.)	
Unit-2: Data Storage and Information Retrieval	Contact Hours: 12
Data Storage: Database Design: Schema, ER Diagrams, Relational Databases (SQL) vs NoSQL (MongoDB), Writing SQL Queries, Functions, Stored Procedures, Indexing and Search Optimization: B+ Trees, Hash Indexes, Data Modeling and Normalization, Client-Server Architecture: Concepts and components. Information Retrieval: Index Construction for Efficient Search, Scoring Models: TF-IDF, BM25, Search Engine Architecture: Indexing and Query Execution, Search Evaluation: Precision, Recall, F1-score, Mean Average Precision (MAP)	



Unit-3: Data Structures, Algorithms, and Processing			Contact Hours: 12
Data Structures: Basics: Arrays, Stacks, Queues, Linked Lists, Advanced Data Structures: Graphs, Trees, Heaps, Memory Models: Associated memory, Hash Maps, Dictionaries. Searching Algorithms: Linear Search, Binary Search, DFS, BFS, Sorting Algorithms: Quick Sort, Merge Sort, Bubble Sort, Heap Sort, Graph Algorithms: BFS, DFS, Dijkstra's Algorithm, Complexity Analysis: Big O notation, Time and Space Complexity			
Unit-4: Data Analysis and Visualization			Contact Hours: 10
Data Analysis: Regression Analysis: Linear, Logistic, Principal Component Analysis (PCA) for Dimensionality Reduction, Canonical Correlation Analysis (CCA), Analysis of Variance (ANOVA) Data Visualization: Basic Visualizations: Histograms, Pie Charts, Box Plots, Advanced Visualizations: Scatter Plots, Bubble Plots, Area Plots, Interactive Visualizations: Waffle Charts, Word Clouds, Visualizing High-Dimensional Data using t-SNE, PCA			
List of Experiment			
Unit-1: Data Collection and Preprocessing			Contact Hours: 6
1. Data Collection from Multiple Sources (Text, Video, Audio, IoT, APIs) 2. Handling Missing Data, Noise Elimination, and Outlier Detection 3. Feature Engineering and Dimensionality Reduction (PCA, t-SNE, LDA, Encoding, Normalization)			
Unit-2: Data Storage and Information Retrieval			Contact Hours: 8
4. Database Design and SQL Queries (Relational Databases) 5. NoSQL Database Implementation and Indexing (MongoDB) 6. Information Retrieval and Search Evaluation (TF-IDF, BM25, Precision/Recall/F1/MAP)			
Unit-3: Data Structures, Algorithms, and Processing			Contact Hours: 8
7. Implementation of Basic and Advanced Data Structures (Arrays, Stacks, Queues, Linked Lists, Trees, Graphs, Heaps) 8. Searching and Sorting Algorithms (Linear, Binary, Quick, Merge, Heap Sort) 9. Graph Algorithms and Complexity Analysis (BFS, DFS, Dijkstra, Big O analysis)			
Unit-4: Data Analysis and Visualization			Contact Hours: 8
10. Regression Analysis and Dimensionality Reduction (Linear, Logistic, PCA, CCA, ANOVA) 11. Basic Data Visualization (Histograms, Pie Charts, Box Plots, Scatter Plots) 12. Advanced and Interactive Data Visualization (Bubble Plots, Area Plots, Waffle Charts, Word Clouds, High-Dimensional Visualization using PCA/t-SNE)			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.



Problem Solving / Coding	5 Python / SQL data-engineering tasks	10	Based on the coding solutions submitted.
Online Technical Quizzes	5 quizzes / simulations on data modelling, ETL & SQL	5	Based on quiz scores.
Micro Project	build a small end-to-end data pipeline	10	Based on Project Report
OR			
Online Courses: NPTEL/Coursera/MooC's/Swayam (As Per Available Hours/Credit's).			
• Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Plagiarism & AI Content Free as Per the Rule.			

Textbooks:

1. Introduction to Algorithms – Cormen, Leiserson, Rivest, Stein. MIT Press, 3rd Edition. ISBN-13: 978-0262533058
2. Database System Concepts – Silberschatz, Korth, Sudarshan. McGraw Hill Education, 6th Edition. ISBN-13: 978-9332901384
3. Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools – Cielen, Meysman, Ali. Dreamtech Press. ISBN-13: 978-9351199373

Reference Books:

1. Data Engineering: A Novel Approach to Data Design – Brian Shive. Technics Publications. ISBN-13: 978-1935504603
2. Python Data Science Handbook: Essential Tools for Working with Data – Joel Grus. O'Reilly. ISBN-13: 978-9352134915
3. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron. O'Reilly. ISBN-13: 978-1492032649
4. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython – Wes McKinney. O'Reilly. ISBN-13: 978-1491957660.

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**PANIPAT INSTITUTE OF
ENGINEERING & TECHNOLOGY**

(Approved by AICTE, New Delhi & Affiliated to Kurukshetra University, Kurukshetra)

AN AUTONOMOUS INSTITUTE

(6th Sem)

Exit

Course



Course Code				Course Title				L	T	P	SL	Credits	
BT-CSE-401A				Neural Networks and Deep Learning				45	-	-	45	90/30 = 3	
								CIE		SEE		Total	
								40		60		100	
Course Outcomes: After Completion of Course, Students will be able to													
CO1	Explain the fundamentals of neural networks, perceptron models, activation functions, and learning paradigms.												
CO2	Apply training algorithms to develop neural network models (BPN, Hopfield, BAM, SOM).												
CO3	Analyze advanced architecture (ART, RNNs, LSTMs, GRUs, Autoencoders) for classification and forecasting.												
CO4	Design and optimize deep learning models using advanced architecture, transfer learning, and modern frameworks.												
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)													
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	1	-	-	-	-	-	-	2	1	-
CO2	3	3	2	2	2	-	-	-	-	-	3	2	2
CO3	3	3	3	3	2	-	-	-	-	-	3	3	2
CO4	3	3	3	3	3	-	-	-	-	-	3	3	3
Avg	3.00	2.75	2.25	2.25	2.33	-	-	-	-	-	2.75	2.25	2.33

Course Outline

Unit: 1 Fundamentals of Neural Networks	Contact Hours: 11
Introduction: Fundamentals of biological neural networks, Basic concepts of artificial neural network: taxonomy of architecture, terminologies, history & applications of neural networks, Perceptron neural network: Single-layer perception, multi-layer perception, Linear separability, XOR problem. Mathematical Foundations: Activation functions: Linear, Threshold, Sigmoid, Tanh, ReLU, Leaky ReLU, SoftMax, Loss functions: MSE, Cross-Entropy, Training Basics: Gradient Descent, Stochastic Gradient Descent, Forward Propagation. Learning Paradigms: Supervised, Unsupervised, and Reinforcement learning.	
Unit: 2 Neural Network Models & Training	Contact Hours: 11
Backpropagation neural network: architecture and training algorithm, Hopfield network structure and training algorithm, Associative memory architecture, training algorithm and applications, Bi-directional Associative Memory (BAM). Competitive & Self-organizing Networks: Kohonen Self-Organizing Maps (SOM), Counter propagation networks	



Unit: 3 Advanced Neural Architectures	Contact Hours:12		
Adaptive Resonance Theory (ART): Basic concepts, ART1, ART2 networks, classification tasks, Special Networks: Neocognitron & Cognitron, Optical neural networks & holographic models Recurrent Neural Networks (RNNs): Vanilla RNNs, Challenges: vanishing, exploding gradient, Applications: time-series forecasting, text processing, Autoencoders: Basic structure, applications in feature learning & dimensionality reduction.			
Unit: 4 Advanced Deep Learning Models and Frameworks	Contact Hours:11		
Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU), Advanced Convolutional Neural Network. Advanced Deep Learning Model Design Principles: Transfer Learning and Fine-Tuning Strategies, Hyperparameter Optimization and Regularization Techniques, Model Compression, Pruning, and Quantization, Deep Learning Frameworks: TensorFlow, Keras, and PyTorch, Edge AI and TinyML Applications			
TW/SL: Total: 45 Hrs. - (All Activities Must be Related to Subject)			
Activity	Description / Quantity	Hrs.	Evaluation
Assignment Writing	05 Assignments	20	Based on the assignments submitted.
Problem Solving / Coding	5 PyTorch / TensorFlow implementation tasks	10	Based on the coding solutions submitted.
Video Based Learning	video explaining a DL model	10	Based on Summary Report.
Poster / Chart / PPT Preparation	Poster / chart / PPT on a key or trending topic of the course, followed by a short presentation	5	Based on poster / chart quality and presentation skills.
OR			
Online Courses: NPTEL/Coursera/MooC’s/Swayam (As Per Available Hours/Credit’s).			
• Certificate & Report Mandatory Before End of Semester. • If Any Student Complete This Task, then He/She Will be Exempt from above. (Minimum 45 Hrs./03 Credit)			
All Reports Must Be Pilgrims & AI Content Free as Per the Rule.			

Textbooks:

1. Li Min Fu, "Neural Networks in Computer Intelligence", McGraw-Hill, Inc. 2012.
2. S N Sivanandam, "Neural Networks using MATLAB 6.0", TMH, 4th. Reprint 2015.
3. S N Sivanandam, "Principles of Soft Computing", 2nd. Edition, Wiley, Reprint 2014.

Reference Books

1. Freeman J.A. & D.M. Skapura, "Neural Networks: Algorithms, Applications and Programming Techniques", Addison Wesley, Reading, Mass, 2014.
2. Deep Learning (Ian J. Goodfellow, Yoshua Bengio and Aaron Courville), MIT Press, 2016.
3. Deep Learning with Python: A Hands-On Introduction by Ketkar, Apress



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Course Code:		Course Title								L	T	P	Credits		
BT-CSE-473 A		Summer Internship -III								-	-	--	2		
										CIE		SEE		Total	
										50		50		100	
Course Outcomes: After Completion of Course, Students will be able to															
CO1	Technical and managerial skills learned during the internship to real-world or institutional projects.														
CO2	Gain exposure to modern technological tools, practices, and innovations relevant to the student’s domain.														
CO3	Develop professional skills documentation, communication, teamwork, and problem-solving through hands-on training and observation.														
CO4	Produce a comprehensive internship report and presentation demonstrating the application of theory to practical tasks, originality, and analytical understanding.														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	3	2	2	3	2	–	–	1	2	–	2	3	2	
CO2	2	3	2	2	3	2	–	–	–	2	–	2	2	3	
CO3	2	2	2	2	2	–	–	–	3	3	2	2	1	2	
CO4	2	2	1	2	2	–	–	–	2	3	1	3	2	2	
Avg	2.25	2.5	1.75	2	2.5	1	–	–	2.5	2.5	1.5	2.25	2	2.25	

B.Tech. student will be required to attend Inter/Intra Institute Summer Internship-III of 6 weeks duration at the end of 6th semester (before the commencement of 7th semester) and with the following objectives:
Provide possible opportunities to learn, understand and sharpen the real-time technical / managerial skills required at the job.

Exposure to the current technological developments relevant to the subject area of training.

The students are required to be involved in one or more of the followings Inter/ Intra Institutional activities viz; Training with higher Institutions; Soft skill training organized by Training and Placement Cell of the respective institutions; contribution at incubation/ innovation /entrepreneurship cell of the institute; participation in conferences/ workshops/ competitions etc.; Learning at Departmental Lab/ Tinkering Lab/ Institutional workshop; Working for consultancy/ research project within the institutes and Participation in all the activities of Institute's Innovation Council, eg; IPR workshop/Leadership Talks/ Idea/ Design/ Innovation/ Business Completion/ Technical Expos etc. The activity or set of activities shall, in each case, be recommended by Departmental Training Coordinator/Incharge, if any, and/or Class In-charge of the IIIrd year and approved by Head of the Department.

A student will write daily reports which will cultivate the habit of documentation of daily observation. After completion of Internship, the student shall prepare a comprehensive report to indicate what he has observed



and learnt in the training period. The student may contact Training/Internship In-charge/ Faculty Mentor to assign special topics and problems and should prepare the final report on the assigned topics. Daily diary will also help to a great extent in writing the Internship report since much of the information would already be incorporated by the student into the daily diary.

Documentation Requirements:

- i. Daily Diary: Record of day-wise activities and observations.
- ii. Final Report: Includes all tasks performed, observations, insights, and analysis of the internship project/activity.
- iii. Special Topics/Problems: Assigned by Faculty Mentor or Departmental Internship Coordinator, if applicable.

The Internship report will be evaluated based on following criteria:

- i. Originality (A signed certificate to that effect to be attached by the student in his/her internship report)
- ii. Adequacy and purposeful write-up.
- iii. Organization, format, drawings, sketches, style, language etc. Variety and relevance of learning experience.
- iv. Practical applications, relationships with basic theory and concepts taught in the course.

This Internship will be evaluated based on a presentation made by the students by the following committee as per guidelines stipulated in the Ordinance

- (i) Head of the Department
- (ii) Coordinator of Internship
- (iii) Class In- Charge